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OIPK VALIDATION REPORT

Executive Summary

Overview: This report critically evaluates the *Orthogonal Implosion-Photon Kubus (OIPK)* framework, a speculative cosmological model positing an unusual spacetime structure. In OIPK, the universe's geometry is envisioned as a **cube-on-vertex (diamond) configuration** of spacetime with **vertical 2D time-slices** that collapse ("implode") downward, while **horizontal light-like flows** propagate orthogonally across these slices at the speed of light *c*. Additionally, **12 symmetric directions** (along the edges of the cube/diamond) produce recursive mirror images, suggesting a fractal self-similarity. The model further integrates **consciousness** as a process moving horizontally with light (thus orthogonal to the vertical implosion), and it attributes our perception of an "arrow of time" to an asymmetry introduced at the point of maximum cosmic expansion (a proposed mirror/turnaround point). Finally, the OIPK framework intriguingly links the **Big Bang** to a *"white hole"* (upper vertex of the diamond) and black holes to the lower vertex, implying a black-hole-to-white-hole correspondence. This report synthesizes evidence from theoretical physics, cosmology, mathematics, and neuroscience to assess each of these bold claims, identifies points of contact with established science, and outlines challenges and testable predictions.

Multiple Temporal Dimensions: A central novelty of OIPK is the introduction of effectively two time-like directions: a *vertical implosion time* (governing the collapse of 2D slices) and a *horizontal photon time* (governing the speed-of-light information flow). Notably, mainstream physics largely assumes a single time dimension for consistency with relativity. However, the idea of **more than one time dimension** *has* appeared in theoretical physics – for example, in Itzhak Bars' "two-time physics" models which embed our 4D world in a higher-dimensional space with two timelike coordinates [1]. Bars argued that a 2T approach in *d+2* dimensions yields a more symmetric description projecting down to the usual 1T physics in *d* dimensions [1]. Similarly, F-theory (an extension of string theory) employs a 12-dimensional spacetime with metric signature (10,2), effectively introducing a second time dimension [2]. These theories are speculative, but they demonstrate that *mathematically* consistent frameworks with multiple time-like directions are conceivable. That said, they require extra constraints (gauges or embedding conditions) to avoid the pathological non-causal behavior that additional time axes can introduce. Indeed, an important caveat comes from Max Tegmark's analysis: in a world with two time dimensions, the evolution of physical systems would be largely unpredictable and chaotic (partial differential equations become *ultrahyperbolic*), potentially preventing stable matter and intelligent life [3]. For example, protons and electrons could decay into higher-mass states unless constrained to effectively one time dimension [4]. This is a major theoretical challenge to OIPK's assumption of orthogonal times. **However,** certain models show that if initial data are specified on a carefully chosen mixed-type hypersurface with appropriate constraints, deterministic evolution *is* possible even with multiple time coordinates [5]. The **ultrahyperbolic equation** (a wave equation with two time variables) has a well-posed initial value formulation if one imposes a nonlocal constraint on the initial data and evolves along the additional time dimension [5]. This suggests that OIPK's two "times" might be made mathematically consistent by constraining how the vertical and horizontal dynamics couple, possibly via a special initial surface or gauge condition. In summary, the notion of orthogonal temporal foliations finds some precedent in two-time physics and certain anisotropic gravity theories, but it faces significant stability and causality hurdles. The OIPK model would need to demonstrate how it avoids the instabilities Tegmark and others predict (perhaps via hidden symmetries or conservation laws akin to Bars' 2T-physics approach). One promising connection is to **Hořava–Lifshitz (HL) gravity**, a quantum gravity theory that *does* posit an explicit preferred time direction: HL gravity treats time and space with different scaling (breaking Lorentz symmetry at high energies) and uses a fixed foliation of spacetime [6]. In HL gravity, time is "more fundamental" and at short scales the speed of light effectively becomes infinite, restoring ordinary relativity only at large scales [7] [8]. This framework shows that having a distinguished time coordinate (like OIPK's vertical implosion parameter) is not absurd, and that relativity can emerge as an approximation. Likewise, **causal dynamical triangulations (CDT)** in quantum gravity explicitly build spacetime from slices separated by a preferred discrete time step. In CDT, spacetime is a stack of successive spatial slices (each a 3D triangulated manifold, often taken as S^3) glued by timelike simplices [9]. A global "proper time" coordinate labels the slices [9], much as OIPK assumes a global implosion ordering. Importantly, CDT's use of a preferred foliation does not appear to spoil continuum general covariance in the phases where semiclassical behavior emerges [10]. This gives hope that a slight anisotropy between horizontal (spatial/light) dimensions and a vertical (implosion) dimension might be reconciled with physics as we know it, at least in an effective sense. OIPK might be formulated as a Bianchi-type cosmology (specifically a Kantowski–Sachs or Bianchi I metric) where two scale factors – one for the horizontal plane and one for the vertical direction – evolve differently with a common time parameter. Such anisotropic cosmological solutions are well-studied and do not violate any fundamental principle, although they must meet observational bounds on anisotropy. In summary, **existing theory provides partial support** for the idea of a structured, two-rate time evolution: it is not standard, but analogies (two-time physics, HL gravity, CDT) and anisotropic metrics show how it *might* be implemented. The key will be to formalize OIPK's metric and dynamical laws in a way that preserves causality and yields the known 4D physics in appropriate limits.

Cube-on-Vertex Geometry & 12-Directional Symmetry: OIPK's geometric premise is that spacetime has the topology or tessellation of a cube balanced on a vertex – essentially an *octahedral* symmetry if one looks at it as a diamond shape. This implies 12 preferred directions (along the edges of the cube). No **cosmological standard model** adopts an actual cubic lattice of spacetime; the universe is usually assumed isotropic. However, discrete or lattice models of spacetime have been explored. In Regge calculus and lattice quantum gravity approaches, spacetime can be approximated by gluing simple polytopes; while a *cubic* lattice is not common (simplexes are preferred for isotropy), one can certainly construct toy cosmologies with cubic lattices [11] or other regular tilings. For example, cosmologists have studied an "Einstein–de Sitter universe" with matter at the nodes of a regular cubic lattice as a way to model inhomogeneities [11]. More relevant to OIPK is the concept of **cosmic topology**. It is possible the universe's spatial section is not infinite or simply connected; a famous model by Luminet *et al.* suggested the universe might be a *Poincaré dodecahedral space* (a spherical space tiled by 12 pentagonal faces) to explain certain anomalies in the cosmic microwave background (CMB) [12] [13]. In that model, one would see a kind of "hall of mirrors" effect, where the sky's patterns repeat according to the face identifications. While not a cube, a dodecahedral topology introduces a 12-fold symmetry in the sense of 12 faces. Intriguingly, if one stands at the center of a cube and looks toward the 6 faces (±x, ±y, ±z directions), or along the 4 major diagonals (through opposite vertices), or along the 12 directions of the edges, each set yields symmetrical orientations. **OIPK emphasizes the 12 edge directions**, which are intermediate angles (45° relative to axes). This specific symmetry does not correspond to a common cosmological model, but it resonates with ideas in quasicrystals and fractals. Notably, **quasicrystal patterns** in 2D often have 10-fold or 12-fold rotational symmetries that are "forbidden" in periodic crystals [14]. By analogy, one might imagine a quasicrystalline order to cosmic structures. In fact, a *recent analysis (2025)* by O. Basta applied **quasicrystalline 12-fold tilings** to astrophysical data [15] [16]. Basta found that overlaying a **dodecagonal (12-fold) inflation tiling** on the spiral galaxy NGC 3982 aligned strikingly well with observed features: the galaxy's central supermassive black hole and star-forming knots lay on the lattice's predicted nodes [17] [18]. This suggests nature *can* exhibit 12-fold symmetric arrangements in gravitational systems, albeit this is a speculative and emergent pattern rather than a fundamental law. Moreover, the study extended 12-, 10-, 16-, 20-, and 24-fold quasicrystal tilings to the large-scale structure (in particular the Boötes void) to see if hidden order might lie in the distribution of galaxies and dark matter [19] [20]. While the 20-fold pattern gave the best fit for that void (hinting at decagonal order), the 12-fold pattern was among those tested [21]. These quasicrystal analogies support the **fractal self-similarity** aspect of OIPK: in quasicrystalline tilings, smaller motifs repeat in rotated orientations, reminiscent of recursive mirroring. OIPK's "12-directional recursive mirroring" evokes a *kaleidoscope*. Indeed, placing mirrors in the configuration of a rhombic dodecahedron (the polyhedron with 12 rhombic faces, which is the dual of the cube) produces an optical holoscope that generates repeating images in symmetric directions [22]. This is literally a visual analog of OIPK's idea – the contents of one cell are reflected into 12 angular copies, creating a fractal array of images. In mathematics, the **Coxeter reflection groups** for polyhedra can formalize such constructions. The cube/octahedron symmetry group (denoted O_h) has 24 rotational symmetries; including reflections doubles that. Iterating reflections through intersecting mirror planes can tile space (or the inside of a sphere) with repeated patterns, a technique used in constructing some fractals and sphere packings. While OIPK is not explicitly a tiling of 3D space (it seems to regard the 12 directions as possible pathways or "mirror channels" for information or influence), one could imagine that the state of each 2D time-slice is *copied* or *influenced* along those directions, yielding a self-similar structure at different scales. This resonates with certain fractals like the **Menger sponge** or recursive octahedra, but with the specific number 12 instead of 4 or 8. If each "edge mirroring" in OIPK produces a scaled-down replica of spacetime geometry in 12 orientations, we can estimate the effective fractal dimension D by solving $12 \times r^D = 1$ (where r is the similarity scale factor for one iteration). If, say, each recursion scales lengths by 1/2, then $12 \times (1/2)^D = 1$ gives $D = \log_{1/2}(1/12) \approx 3.584$. This is purely speculative since OIPK has not defined the scale factor, but it underscores that a high fractal dimension close to 4 could result, meaning the structure nearly fills 3D space. In summary, while standard cosmology is highly symmetric (3D isotropy, no preferred directions) and OIPK's *specific* 12-edged diamond structure is novel, there are parallel ideas: cosmic topology with finite symmetry groups, quasicrystal analogies on galaxy distributions, and mirror symmetry fractals. These provide **partial validation** that having a discrete symmetry or repeating pattern is not implausible—though no evidence yet points to a literal cube/octahedral axis in our universe. Notably, CMB studies have looked for unusual symmetries; apart from a tentative "quadrupole-octopole alignment" (the so-called CMB **"Axis of Evil"**), no clear 12-fold pattern has been found. OIPK's prediction of 12-fold correlations in CMB or large-scale structure remains unconfirmed but is testable (one could search CMB maps for a dodecagonal correlation function or attempt to fit quasicrystal templates as was done with galaxies).

2D Slices, Holography, and 3D Reconstruction: OIPK asserts that each instant of the universe is like a *2D upright slice*, akin to a planar holographic film or photographic negative. The horizontal light flow ($\leftarrow\rightarrow$) "illuminates" these slices, presumably giving rise to the experience of a 3D world. This sounds metaphorical, but we can attempt to map it to known physics. The **holographic principle** in physics indeed says that the information content of a volume of space can be encoded on a lower-dimensional boundary (typically, one less dimension). Usually, this is applied to a *closed surface* encircling a volume (e.g. the surface of a black hole encodes the interior information, with entropy $S = A/4L_P^2$). Here, OIPK suggests internal *time* slices act like holographic screens. One is reminded of **Bousso's covariant holographic bound**, which allows one to construct *light-sheets* from any surface and states that the entropy passing through that light-sheet is $\leq$ area/4 in Planck units [23] [24]. For an "upright" time-slice (which we can picture as a 2D spatial cross-section at a given moment τ), one could imagine firing light horizontally through it; that light-sheet would carry information about the slice's state. Bousso's framework even notes that the entropy bound is time-symmetric and hints at an underlying reversible information count [25] [24] – an intriguing parallel to OIPK's idea of a two-way time perception. While the **holographic principle** is usually thought of in terms of surfaces at infinity or event horizons, there's nothing forbidding the idea of stacking multiple holographic "films" through the volume. In fact, Erik Verlinde's **entropic gravity** proposal treats *every* equipotential surface as a holographic screen encoding information about the interior mass distribution [26] [27]. One could imagine discretizing a spatial volume into a foliation of thin layers, each carrying bits that holographically project to build up the 3D image. This is analogous to medical **tomography**: a 3D object can be reconstructed from many 2D scans. In computed tomography (CT), a series of 2D X-ray images at different angles allows computation of the 3D density distribution. In *light-field photography*, an array of slightly different 2D images (captured by shifting the viewpoint or using a micro-lens array) permits refocusing and depth reconstruction. OIPK's horizontal light might similarly integrate successive 2D slices into a volumetric perception. Information-theoretically, if each 2D slice carries S bits (perhaps S scaling with area as in holography), the total information available to reconstruct a segment of 3D space would be the combination of many slices. How many slices N are needed to reconstruct a 3D "frame" of perception? In computer vision, as few as two distinct views (stereo pair) can yield depth via parallax, but with ambiguities; more viewpoints (or continuous motion parallax) improve fidelity. Humans effectively use a continuous stream (~60 frames per second from micro-saccades and eye movements) integrated over ~0.1–0.3 seconds, i.e. on the order of *10–20 frames*, to build a stable 3D understanding in the brain. Interestingly, OIPK postulates on the order of 10^{42} slices integrated over 100–300 ms – this is essentially the number of *Planck time slices* in a typical conscious moment (since one second is ~ 10^{44} Planck times). It seems OIPK is conjecturing that reality at the Planck scale consists of an enormous stack of 2D "bits" that our consciousness somehow sums up. There is a known conjecture by Wheeler: "it from bit," suggesting spacetime could be built from binary information on tiny scales. If indeed each Planck interval yields a 2D "frame" of reality, then conscious perception might be an aggregate that only refreshes at ~10 Hz (100 ms per update), giving the illusion of continuity. This is *broadly consistent* with neuroscience: the **critical flicker fusion frequency (CFF)** for humans is around 50–60 Hz under photopic conditions, meaning flashes faster than ~20 ms apart merge into a continuous perception [28]. Many animals have different CFFs (flies >200 Hz, some mammals ~80 Hz) indicating variations in integration windows [29] [30]. Humans can detect separate events about ~10–20 ms apart, but the **"specious present"** – the window of integrated consciousness – is typically a few tenths of a second. Empirical estimates of the specious present indeed range ~100–300 milliseconds [31] [32]. Within this window, stimuli are bound together into a unitary experience. For instance, if audio and video are offset by more than ~100 ms, the brain perceives them as unsynchronized [33]. This ~0.1 s integration domain corresponds to the brain's known 4–10 Hz oscillatory cycles (theta/alpha waves) and the **P300** event-related potential (which peaks ~300 ms after a salient stimulus) [33]. OIPK's assertion that consciousness integrates ~10^42 "slices" in ~0.1–0.3 s thus coincides with the idea that on the Planck scale, those slices accumulate into one "frame" of awareness. We might view each 2D slice as a **Holographic frame** of the universe, and consciousness as a projector lamp that shines through a stack of frames to produce a moving picture. This is metaphorically similar to the old movie reel: each frame is static, but illuminated in rapid succession they create motion. One can ask: why *2D* time-slices and not 3D? Perhaps OIPK is invoking the **holographic principle** to assert that the fundamental description of each instant is 2-dimensional (like a hologram), which when lit by information flow yields the emergent 3D reality. Leonard Susskind's original holography conjecture did tie 3D spatial volume to a 2D surface data [24]; OIPK internalizes that idea by placing the 2D surfaces *inside* the universe, layered in the time direction. This is an intriguing twist – typically holographic duality in AdS/CFT or similar has the 2D (or lower-D) boundary at infinity or a black hole horizon, not literally slicing through the bulk. However, recent holographic formalisms consider **"finite holographic screens"**: codimension-1 surfaces at finite radius that maximize entropy for given enclosed volume (e.g. in de Sitter space or cosmology, one can define trapping surfaces or apparent horizons as possible holographic screens) [34] [35]. It's conceivable that each OIPK slice is such a screen for the part of spacetime above it. If so, one could attempt to derive a "slice holography" formula: e.g. the information in a layer of thickness $\Delta\tau$ might relate to the area of that slice. OIPK also brings to mind **3D displays** like volumetric or light-field displays, which use multiple 2D images/sheets to create the illusion of depth. For example, a simple volumetric display might use a rapidly spinning 2D screen that displays different images at different z-positions, which the eye integrates into a 3D image. In an even closer analogy, a **light field** captured by many cameras (e.g. a 4×4 grid of cameras) can be thought of as capturing many 2D projections of a 3D scene; algorithms can then reconstruct the 3D scene or refocus images. If the universe's true state is 2D slices, then the myriad horizontal light rays crisscrossing might act like a plenoptic camera, encoding 3D information in multiple 2D projections. A rough calculation: if each 2D slice has entropy proportional to area ($S \sim A/4L_P^2$ as per holography), then N such slices could encode up to $N \times A/4$ worth of information (assuming independent content on each). To reconstruct a 3D region of volume $V \approx A \cdot L$ (thickness L), one might need on the order of L/L_P slices (if each slice is a Planck length apart). That would give $S_{\text{total}} \sim (L/L_P)\cdot (A/4L_P^2) = \frac{A L}{4L_P^3} = \frac{V}{4L_P^3}$, which interestingly is the maximum entropy of a **3D volume V filled with ~one bit per Planck volume**. This is the expected order of magnitude for holographic capacity (though gravity actually suggests far fewer bits – only area/Planck^2, not volume/Planck^3). So unless the slices' info is redundant or constrained, stacking area-encoders can overcount degrees of freedom. It may be that the 2D slices in OIPK are not independent "screens" but correlated by the dynamical law (the vertical implosion). Thus, a consistent formalization might be that the 3D state at any given time arises from a *cumulative projection* of many past 2D slices, rather than a simple sum of independent bits. This is speculative, but at least concepts from **holography** and **tomography** lend some plausibility to the notion of building 3D reality from 2D data slices via light. A concrete test could be to simulate a toy model: for instance, take a 3D object, produce a series of 2D slices, then try to reconstruct the object by illuminating those slices with shifting 2D patterns (analogous to horizontal light rays). This could be done in a lab with diffraction/holographic optical elements. If successful, it would provide a nice metaphor for OIPK's mechanism.

Asynchronous Vertical vs Horizontal Dynamics: In OIPK, the vertical implosion proceeds at some rate v_{imp} which is not necessarily equal to *c*. This means the "creation or collapse of space" vertically could happen slower or faster than light can cross a slice. This is akin to having **two clocks** in the system: one measures the rate of succession of slices (vertical proper time, perhaps), and one measures the standard time along each slice (horizontal light travel time). Are there known physical systems with two time-like parameters? Directly, as discussed, multiple time dimensions are rare. However, **parametrized theories** often have an external evolution parameter separate from coordinate time. For instance, the Wheeler–DeWitt equation in quantum gravity $H\Psi = 0$ has no explicit time; instead, one can choose an internal degree of freedom (like the universe's scale factor or a scalar field) as a *clock* to parametrize evolution of others. If OIPK's vertical dimension τ is something like the scale factor (which changes as the universe expands or contracts), one could treat τ as an evolutionary parameter for the state on each horizontal slice. This resembles **Julian Barbour's** perspective: Barbour argues time is simply a measure of change of the whole configuration (he calls his timeless configuration space *Platonia*). The OIPK vertical direction (implosion from expansion to contraction) could be playing the role of Barbour's "time" – essentially indexing different 2D configurations ("Nows"). Barbour's recent *Janus point* theory posits a unique turning point (of minimum entropy or complexity) from which time runs in opposite directions [36] [37]. If OIPK identifies the maximal expansion as a special mirror, it aligns conceptually with Barbour's idea (except OIPK suggests a maximal *expansion* rather than minimal *size* might be the Janus mirror – in a closed universe, those coincide, since maximal expansion is a turnaround to contraction). Another relevant framework is **two-clock cosmologies**. Some brane-world scenarios or bimetric theories allow different "speeds" for propagation in a hidden dimension vs our dimension. **Variable speed of light (VSL)** cosmologies, for example, imagine c was higher in the early universe to solve horizon problems. This is effectively making the light-crossing horizontal time shorter relative to expansion time, for a period. If v_{imp} were extremely high in the early universe (rapid expansion of slices) compared to c, one gets something inflation-like (space expands/implodes faster than light moves across it). Conversely, if v_{imp} is usually slower than c, then within one slice horizon-size, light has time to interact and maintain causal contact, which is good. **Mathematically,** OIPK might be represented by a metric with anisotropic scale factors:

$$ds^2 = -c^2 dt^2 + a^2(\tau)\,(dx^2+dy^2) + b^2(\tau)\,dz^2,$$

where z is the vertical coordinate and τ is some function of z or the proper time along z-lines. Here t corresponds to horizontal time (with speed c) and τ is like cosmic time controlling the scale factors. If one treats τ as a second time coordinate, the metric could formally be extended to $ds^2 = -c^2 dt^2 + b^2\,d\tau^2 + a^2(\tau)\,(dx^2+dy^2)$, which has two time-like directions if $d\tau$ is timelike. But more likely, τ is a parameter not present as an independent coordinate in the line element, just the label for how a and b change. The universe might start at $\tau=0$ (Big Bang/white hole at the upper vertex), expand $a(\tau)$ up to $\tau=\tau_{\max}$ (maximum expansion, midpoint of diamond), then contract back down to τ_{final} (Big Crunch/black hole at lower vertex). Meanwhile, $b(\tau)$ might measure how the vertical separation between slices behaves – possibly $b(\tau)$ shrinks to 0 at both endpoints (a "cubic" geometry that pinches at top and bottom). The OIPK's "implosion" suggests $b(\tau)$ decreases as τ increases (if we take vertical volume collapsing). If $v_{\text{imp}} = d(b)/d(t)$ or something, we could compare it to c. Without a detailed law, we consider analogies: In **ADM formalism** of general relativity, one splits spacetime into slices and has a lapse function $N(dt)$ and shift vector. Different lapse choices mean the coordinate time runs differently relative to proper time on slices – essentially giving freedom to "move" through slices at different rates. So OIPK could be seen as picking a certain lapse (one that is not trivial) meaning the proper time of implosion is not the same as the coordinate time t. In simpler terms, imagine a cartoon: each 2D slice "dissolves" after some time Δt_{dwell}, and the next one takes over. If Δt_{dwell} is much smaller than light crossing time of the slice, then light wouldn't fully propagate before the slice moves – chaotic. If it's much larger, then effectively things settle to near-equilibrium on each slice before the next comes – that's slow implosion. The balance between these could be parameterized by a dimensionless ratio $\epsilon = v_{\text{imp}}/c$. If $\epsilon \ll 1$, vertical collapse is slow relative to light, yielding quasi-static slices (like an adiabatic evolution). If $\epsilon \sim 1$ or >1, then there's a strong departure from relativity on large scales – possibly detectable as Lorentz violation or anisotropy. We know experimentally that the universe on large scales is **Lorentz invariant** to high precision (e.g. no observed anisotropy in speed of light across different directions to 10^{-15} or better). So OIPK must either have ϵ effectively *zero* (no real breaking of simultaneity, making it trivial), or it must only manifest in contexts that haven't been probed (maybe only at Planck scales or in the structure of black holes). *If* OIPK's vertical direction is related to cosmic expansion, there is an interesting observation: The universe's expansion rate (the Hubble parameter H) provides a natural velocity scale when multiplied by distance. At some scale (the Hubble horizon), recession speed equals c. Right now $H_0 \approx 70$ km/s/Mpc, so at about 4300 Mpc, space recedes at c. During inflation or other phases, space can expand faster than light over some scale. This is not a violation of relativity because it's the metric changing, not motion through space. If we interpret v_{imp} as the *speed of spatial collapse/expansion*, then indeed general relativity allows v_{imp} to exceed c under certain conditions (like inflationary expansion). So OIPK could conceptually accommodate $v_{\text{imp}} \ne c$ without contradiction, provided it's framed as a metric expansion rate, not a signal speed. **The challenge** is ensuring that no information or physical influence travels superluminally in a way that breaks causality. If horizontal light is the only carrier of information, and vertical implosion is more like a background metric change, causality might still hold. One might draw an analogy to an elevator moving (vertical implosion) while light signals move inside it (horizontal). As long as the elevator's speed is below c, no big issues; if above c, it's like a warp drive concept. HL gravity again provides insight: it permits different propagation speeds for gravitons at high energy (potentially infinite speed, as noted) [8], yet it aims to remain causal with respect to a preferred foliation. If OIPK has a preferred "aether" frame given by the slices, it might break global Lorentz invariance but preserve a consistent causal order via that frame (somewhat like a "Newtonian" time in the background). This *would* be testable: e.g. look for slight violations of Lorentz invariance or differences in photon vs graviton speed. The neutron star merger GW170817 ruled out large differences between gravitational wave speed and light speed (they arrived within 2 s over 120 million ly, so $|c_{GW}-c|/c < 10^{-15}$) [38]. This severely constrains any model (like HL gravity or OIPK) that might allow anisotropic propagation—any deviation must be at Planck-suppressed levels. In summary, physics does allow one to separate “evolution rate of the universe” from “light propagation rate” in formalism, but observations insist they *appear unified* to us at low energies. OIPK would need to hide its asynchronous dynamics perhaps in realms we haven’t observed (early universe, interiors of black holes, or the Planck scale).

12-Directional Recursive Mirroring (Fractality and Symmetry): As touched on, the presence of a 12-fold symmetry evokes parallels to quasicrystals and possibly to certain symmetry groups. The cube (octahedral) symmetry group is of order 24 (rotations), but OIPK explicitly calls out 12 mirror directions along edges. This is reminiscent of a **kaleidoscope of mirrors** meeting at 120° angles (since at each cube vertex, three edges meet). Indeed, a triangle of mirrors can create multiple reflections: for example, a three-mirror system with angles tuned appropriately yields a finite symmetry group (if angles are submultiples of 180°). The cube’s edges correspond to an angle of 90° between mirror planes (if one places mirrors along the internal angle bisectors of faces through an edge). A **Wythoff construction** in polyhedra uses mirror planes to generate the uniform polyhedra via reflection groups. If OIPK’s 12 directions act effectively like mirror lines, one could be describing a *group action* on the space of states. Perhaps the state of each 2D slice is copied into 12 symmetric states, which then interfere or combine (like images in a hall of mirrors). This could generate a self-similar or repeating structure both in space and across scales. There are known **fractals with octahedral symmetry** (e.g., one can recursively inscribe smaller octahedra at the vertices of a larger one, creating a fractal). The *Sierpinski octahedron* (sometimes called a 3D Sierpinski gasket) is made by removing smaller octahedral holes, leaving a fractal of dimension ~2.33. The *Menger sponge* has cubic symmetry and dimension ~2.73. OIPK’s fractal might be of a different sort since it implies a perhaps *nonlinear feedback* (mirroring might involve physical fields reflecting back and forth). This could bear some relation to **spin networks** in loop quantum gravity, where each node with multiple links can be thought of as a polyhedron. A 6-valent node, for instance, can correspond to a quantum geometry with 6 faces – essentially a hexahedron (cube) in a certain state [39] [40]. If nature’s fundamental spin-network nodes were 6-valent with cubic geometry, the interconnections might encode 90° or 120° angles – that’s speculation, but it’s curious to note. In any case, a key testable prediction from the 12-direction hypothesis is **anisotropy**. If these directions have any fixed orientation in space, we might detect e.g. a 12-fold modulation in cosmic ray arrival directions, or a preferred alignment of polarization of distant galaxies, etc. None have been reported (we typically see 2-fold dipoles or maybe quadrupole alignments, but not 12). However, if the 12 directions are not *global* but rather local to the “frame” of each imploding region (like each point sees its own local 12 directions, perhaps related by rotations), then global anisotropy might be washed out. It could be more subtle, like the way a crystal’s domains average out. Alternatively, OIPK might only reveal its 12-fold symmetry in special conditions (maybe near black holes or at cosmic turnaround). The fractal mirroring might also hint at a solution to the **information redundancy**: if each slice’s info is mirrored 12 times, perhaps the model ensures that no information is truly lost (lost in one edge reflection but captured in another). This could relate to error-correcting codes in holography, where information is stored in multiple ways (the AdS/CFT correspondence indeed shows how local bulk information is redundantly encoded in many degrees of freedom on the boundary). It is conceivable that the 12 recursive mirrors in OIPK serve as a built-in error-correcting scheme for the universe’s information, ensuring unitarity. While these ideas are highly theoretical, **supporting evidence** for fractal structures in the universe does exist in some contexts. The distribution of galaxies up to ~100 Mpc shows a complex web with filamentary structures that some have described as fractal-like (though on larger scales it smooths out). Additionally, the pattern of voids and clusters has sometimes been analyzed with fractal dimensions (~2 < D < 3). If OIPK is correct, we might expect to see **self-similar clustering** at multiple scales, potentially with some signature of 12 directions in the correlation function. The research by Basta already mentioned found that applying fractal substitution rules (Fibonacci/Pell sequences) with 12-fold symmetry could highlight structures in galaxy maps [41] [42]. This kind of analysis is just beginning; if validated, it would lend credence to the idea of quasicrystalline order in cosmology, dovetailing with OIPK’s premise.

Consciousness Integration & Time Perception: One of the boldest claims of OIPK is that consciousness is somehow *embedded* in the horizontal light flow, perpendicular to the vertical collapse of slices. In other words, our conscious awareness travels with the photons (metaphorically) and is thus not tied to the implosion time axis. This is an attempt to explain why we sense time as flowing in one direction (we move along the slices with the light) even though the underlying structure might allow bidirectional or more complex time. It also provides a mechanism for why we integrate a huge number of fine-grained physical events into a coherent "now." We've already noted that the **specious present ~ 100–300 ms** is an empirically well-established window [32] [33]. Within that window, the brain likely gathers and synchronizes information across different neural processes. In neuroscience, theories like **Global Neuronal Workspace (GNW)** by Dehaene and Baars propose that consciousness occurs when information is globally broadcast across the brain's networks, allowing different modules to integrate the data into a single experience [43] [44]. This broadcasting has been linked to oscillatory activity in the beta and gamma range (~30–100 Hz) which can enable synchrony across distant brain regions. Notably, Baars' **global workspace theory** analogizes consciousness to a *theater spotlight* that illuminates content for many "audience" processes [45] [46]. The spotlight idea is very similar to OIPK's horizontal light illuminating slices. In the brain, the "spotlight" might correspond to attention focusing on certain neural representations and thereby igniting a burst of synchronized firing that is distributed widely (the ignition being perhaps 100 ms in duration). The GNW model indeed suggests that conscious broadcast events are discrete and on the order of ~100 ms long, alternating with periods of unconscious processing [47] [48]. Baars and others have explicitly noted that the *100 ms timescale* appears as a fundamental rhythm in conscious brain physiology [33], which matches the flicker-fusion and integration limits earlier discussed. This is supportive of OIPK's notion that ~10^42 sub-events (Planck frames) could be merged by a mechanism (like the brain) into one moment – because the brain itself seems to slice time into ~0.1 s frames. Moreover, OIPK's assertion that consciousness flows horizontally implies that *within each slice, consciousness can sample information across space essentially at light speed.* This might correspond to the idea that we build a world-simulation from the light (or other sensory signals) reaching us. It could also be hinting at something exotic: perhaps that the mind is associated with *electromagnetic fields (light)* in the brain, which propagate horizontally (within the 2D plane of present) as opposed to being tied to any "vertical" evolution. Interestingly, a hypothesis by John Cramer (the transactional interpretation of quantum mechanics) imagines waves traveling forward and backward in time to create a coherent event. OIPK could be saying that consciousness only deals with the time-symmetric, *horizontal* wave propagation and not with the irreversible collapse (vertical implosion) except at special boundaries. This might be a stretch, but consider that many brain processes are indeed electromagnetic and thus light-like (though at much lower speeds due to effective refractive indices in tissue). On the other hand, *vertical implosion* might correspond to entropy increase and memory formation – processes which are slower and give time its arrow. If consciousness "dodges" the arrow by sticking to the photon frame, that might be why it feels time-neutral or even why time can seem to slow down or speed up depending on attention (if attention rides on different pathways of neural communication). Beyond speculation, some empirical correlations can be noted: Humans have a maximum temporal resolution as discussed (CFF, etc.), and a minimum reaction time (~200 ms for simple tasks, aligning with P300). This implies our awareness cannot catch events much shorter than tens of milliseconds. Those events remain unconscious unless later inferred. So indeed, consciousness appears to be a *serial* process updated a few times a second. In OIPK, perhaps each conscious update corresponds to scanning one 2D slice thoroughly with horizontal signals. The brain's **binding problem** (how disparate sensory inputs unified) might relate: synchronous oscillations (around 40 Hz) have been proposed as the solution to binding (the *feature integration theory*). These oscillations occur in the horizontal cortical circuits (within and across areas). So one could say that "consciousness runs horizontally" across cortex, integrating features, which agrees with OIPK's phrasing. Meanwhile, vertical processes – perhaps subcortical loops or sequential processing – might set up the next scene (like vertical implosion). Another domain: **Integrated Information Theory (IIT)** by Tononi quantifies consciousness by the amount of information integrated across the system. A highly parallel, recurrent network (like cortex) can integrate a lot of info (~bits equal to effective connections). The number ~10^{42} is astronomically larger than any brain's integrated information (which might be, say, ~bits equal to number of synapses 10^{14} or so). So 10^{42} likely refers to fundamental physical events being integrated. If one imagines each conscious moment is a *quantum reduction* of 10^{42} entangled Planck-scale events (Penrose's Orch-OR theory had a vaguely similar numerical estimate based on gravitational microsuperpositions collapsing at 40 Hz), then OIPK aligns with that kind of thinking. Indeed, Penrose and Hameroff's Orch-OR posits that microtubules in neurons orchestrate a quantum state that collapses ~40 times per second, and they tie 40 Hz to conscious moments. OIPK's numbers could be echoing that idea in a different guise (10^42 Planck ops ~ collapse per 0.1 s). **In summary**, current neuroscience supports several elements: (1) There is a discrete integration time (~0.1 s) for conscious perception [33]. (2) Information is integrated by horizontal (cortical) interactions (brain's connectivity favors within-layer and inter-area connections across the cortical sheet) with global broadcasting (which we could liken to light shining across a plane) [43]. (3) The arrow of time in experience is puzzling – physically, micro laws are time-symmetric, but we feel an irreversible flow. OIPK attributes this to being at a "leading edge" of the implosion, with the mirror at maximum expansion ensuring that what's ahead of us (the future) is a reflection of the past. This is somewhat analogous to **Wheeler–Feynman absorber theory** in EM, where the present is determined by a balance of influences from past and future, which only works in a time-symmetric cosmology (all radiation eventually absorbed). If our consciousness is only seeing one side of the mirror, that could explain time's arrow. It is a bit mystical, but at least one **philosopher** had a similar notion: J. W. Dunne in 1927 proposed that to explain the flow of time, one needs a second time dimension and a higher-order self observing the timeline [49] [50]. He envisioned an infinite hierarchy of time levels and conscious selves, leading to an ultimate observer outside all time [51] [52]. OIPK is not as elaborate, but the idea of consciousness on a separate orthogonal time (horizontal vs vertical) is strikingly close to Dunne's model. Dunne's theory faced logical criticisms (the infinite regress), but it's historically notable that he linked multi-time and consciousness explicitly [49]. Modern neuroscience doesn't invoke multiple time dimensions, but it does differentiate between subjective time (constructed by the brain) and objective time. The "horizontal" consciousness could simply be the brain's subjective time, which can differ from cosmic time (e.g., under adrenaline, one's subjective timeline might slow relative to actual clock time – interestingly, high arousal can make more events seem to fit in a short interval, implying a higher sampling rate of consciousness).

Asymmetric Time Perception & Cosmic Expansion Mirror: OIPK proposes that the **maximum expansion** of the universe acts like a mirror or reversal point for time's arrow. This resonates with certain symmetric cosmologies. For instance, a simple closed FRW universe expands from a Big Bang to a maximum size and then recollapses to a Big Crunch. In such a time-symmetric model, one could imagine that entropy increases during expansion, reaches a peak, then decreases during contraction if the contraction phase is a time-reverse of expansion. However, classical thermodynamics would say entropy would keep increasing (making the contraction different from the expansion). Some have proposed special low-entropy conditions at the turnaround to enforce symmetry. **Conformal Cyclic Cosmology (CCC)** by Penrose is a different twist: it identifies the infinite future of our universe with the Big Bang of a next one via a conformal rescaling [53] [54]. In CCC, what we think of as "end of time" (in an exponentially expanding de Sitter phase) becomes literally the "beginning of time" for a new aeon after a mathematical stretching that makes infinite time finite. This provides a kind of time mirror, though not at a finite time but at infinity. CCC also enforces an arrow of time: each aeon has a low-entropy Big Bang and a high-entropy future; by the conformal trick, the high-entropy future is essentially erased (mass decays away) so that the new Big Bang can start afresh in low entropy [55] [56]. Interestingly, CCC predicts that information is not lost in black holes but ultimately leaks out as a very subtle effect in the next aeon (via gravitational wave imprints etc.) [57] [56]. Penrose even suggested that certain anomalous concentric low-variance circles in the CMB could be evidence of signals from black hole collisions in the previous aeon [58]. Those claims remain controversial – some analyses reported tentative signals, others found none beyond random expectations. But the key point is CCC implements an inter-cycle *reflection* (in a conformal sense). Another symmetric model is the **Janus cosmology** mentioned: Barbour, et al., suggest time might flow outward in two directions from a minimum complexity point (which could be like a bounce). If our universe were one side and there's another "mirror universe" on the other side of the bounce, that could explain why overall the state at the bounce is very ordered (both universes start ordered and grow disorder away from the bounce in opposite time directions). OIPK's maximal expansion point acting as a mirror is analogous: it is essentially positing a **time-symmetric universe** where the moment of greatest volume is an inflection point. Before that, from Big Bang to expansion max, the arrow of time points forward (entropy increasing). After that, if the arrow *flips*, an observer in the second half would consider their direction forward (toward the final crunch) as increasing entropy for them, but that would look like entropy decreasing from our perspective. This is tricky: it might imply that what we call the far future (approaching the mirror from our side) starts to *reorder* itself, which is not observed – unless no observers exist that deep into the future. Some speculative proposals toy with this: e.g., **"Gold universe"** models (after Thomas Gold) considered a contraction phase where entropy might decrease (with advanced waves setting things up). No evidence for that so far. However, if the turnaround is in our future (billions of years hence), current observations wouldn't easily see its effects except maybe subtle correlations. We do see that the universe's expansion is currently accelerating, as if approaching a de Sitter phase. A true recollapse would require dark energy to eventually dissipate or become attractive; otherwise, we might expand forever. OIPK might need a modification of dark energy – perhaps in OIPK, dark energy is an emergent effect of approaching the "mirror" (some kind of boundary conditions). One interesting coincidence: The acceleration today (dark energy density ~ 7×10^{-30} g/cm^3) is of the same order of magnitude as matter density was at redshift ~ a few – which is often called the **"cosmic coincidence problem"** (why are we at an epoch where dark energy and matter are comparable?). Some theorists suggest this hints we live not too far from the turnaround or some special transition. If maximum expansion were, say, only a few times the present scale, we'd indeed be living near the "mid-point" of the cosmic timeline – which is statistically strange unless selection effects are at play. At present, **observational data** (type Ia supernovae, CMB, BAO) favor an endless expansion ($w \approx -1$, a cosmological constant). But uncertainties in the Hubble constant and growth rates (the **Hubble tension**) leave room for exotic physics. A turnaround could occur if dark energy is not a constant but a field that will eventually decay or become gravitationally attractive. This is not ruled out – phantom energy ($w < -1$) could even lead to a future collapse (after a Big Rip), albeit phantom typically ends catastrophically, not smoothly turning around. Alternatively, some cyclic models (Steinhardt's ekpyrotic universe) had a slow turnaround via a scalar field potential. The **testable prediction** here is: if we are approaching a mirror, there might be signs like unusual isotropy in far-future state, or subtle correlations in the CMB from future effects (very speculative, akin to Cramer's absorber theory requiring a future absorber). One potential observable Penrose pointed out: the **"Weyl curvature hypothesis"** – an initial low entropy (smooth curvature) and a final very high entropy (with lots of black holes, gravitational clumping). If time-symmetry were to hold, the final state should in some sense be as special as the initial. Penrose didn't believe a classical recollapse can do that – hence his move to conformal magic. OIPK doesn't detail the mechanism, but presumably the 12-fold mirror recursion at maximum size does *something* like reset or reflect the degrees of freedom. Could it be that at max expansion, the 12 mirror images of all structures converge and interfere in a way that reduces entropy? This is highly speculative, but mathematically, if the state of the universe has a discrete symmetry, at the symmetry point configurations can become more ordered due to constructive interference. Think of a system that expands and then at some symmetric point, clones of the system overlap – if they overlap in phase, it might "organize" the configuration. Another tenuous but imaginative link: **black hole - white hole pair creation** at a bounce. Smolin's cosmological natural selection (discussed later) envisioned new universes born in black holes [59]. If our universe's bounce connects white holes to black holes, maybe maximal expansion is where these connections line up. This is far-fetched, but the point is, to validate OIPK's time mirror claim, we'd need evidence of time symmetry or a cycling cosmology. At present, we have hints against it (arrow of time is strong, and no reversal observed), but also theoretical discontent with a true one-way model (because of initial low entropy problem). **The existence of a time-symmetric solution** that matches all known physics except at extreme scales would be groundbreaking. *If OIPK is onto something,* one possible signal could be **CMB anomalies** of a specific kind – for instance, Penrose's circles or some *multi-pole alignments reflecting an early slow contraction or new quantum effect near turnaround.* Another might be the **absence of Boltzmann brains**: in an ever-expanding de Sitter, we'd expect eventually occasional freak observer fluctuations. If time somehow ends (or restarts) at a finite horizon, Boltzmann brains can be avoided. If surveys of the far future find no evidence of random fluctuations (hard to test, of course), that might indirectly support a cutoff. Overall, the maximal expansion mirror idea is intriguing but currently sits mostly in the realm of theoretical possibility (Penrose's CCC being a major example, albeit at infinity rather than finite time).

White Hole – Black Hole Connection: OIPK posits that the upper vertex of the diamond (the Big Bang, presumably) is a **white hole**, while the lower vertex (perhaps the ultimate fate or maybe each black hole interior) is a black hole that connects to it. The concept of identifying the Big Bang as a white hole explosion is not new in speculative cosmology. In classical GR, time-reversal of a black hole is a white hole that expels matter but cannot be entered (the opposite of a BH event horizon). Our Big Bang had low entropy and extreme density – superficially like a white hole ejection (except a white hole in GR has a past singularity and is usually considered unstable). **Several proposals connect black holes to new universes**: Lee Smolin's *fecund universes* hypothesis suggests each black hole's core becomes a big bang of a new, causally disconnected universe, with slight parameter changes [59]. That effectively treats black holes as "white holes" for baby universes. While unconfirmed, this idea elegantly offers an explanation for fine-tuning (universes that make lots of black holes get selected) [60] [61]. Another line of thought comes from loop quantum gravity: **Loop Quantum Cosmology (LQC)** finds that quantum effects can replace the classical Big Bang singularity with a bounce – one could interpret the pre-bounce branch as the interior of a black hole from a parent universe. Similarly, **Planck star** models by Rovelli & Vidotto propose that black hole cores stop collapsing at Planck density and rebound into white holes. These white holes would appear after a long delay (due to relativistic time dilation) as sudden explosions of matter (possibly explaining fast radio bursts or certain high-energy cosmic events) [62] [63]. In 2019, researchers even suggested that mysterious cosmic rays or gamma rays might be mini white hole bursts (though no clear evidence yet). The "black hole fireworks" scenario by Haggard and Rovelli provides a concrete mechanism: a black hole can **quantum tunnel** to a white hole state, releasing information in the process [64]. They showed a toy metric where a collapse leads to a white hole with only a small region of spacetime where quantum gravity is important, and that the process is consistent with causality if the timescales are huge [64]. Crucially, in their model, the information trapped in a black hole is not lost – it's emitted when the hole transitions to a white hole and blasts out. This addresses the **black hole information paradox** by allowing information to escape, but not via a "firewall" at the horizon – rather through a late-time quantum transition [65]. OIPK's identification of the entire universe's timeline with a single black-to-white transition is a grand extension of these ideas. Perhaps the *entire Big Crunch* of a previous cosmological cycle became our Big Bang (a single enormous white hole). If our universe recollapses, maybe it will form a black hole that spawns another universe – a cyclic model not in time (like CCC's successive eons) but in a branching multiverse sense. However, OIPK diagrams a single diamond – suggesting one bounce, not an eternal series. It could also be implying a **universal black hole/white hole duality**: every black hole in our universe potentially corresponds to a "pocket" big bang (like a small diamond), or conversely that the Big Bang was a particularly giant white hole whose black hole counterpart might exist "below" our universe (in some higher-dimensional space or prior epoch). If OIPK is true, one might expect some unusual consequences: for example, maybe every black hole in our universe eventually opens into some other region (perhaps accounting for the 12 mirror channels – could those be gateways for information from other holes?). Observationally, how to test this? One idea: **Planck-scale remnant radiation or echoes.** If black holes transition to white holes, there could be gravitational wave **echoes** following black hole mergers (faint repetitive signals after the main ringdown, as proposed by some quantum gravity phenomenologists). LIGO has looked for echoes; tentative hints have been reported in some analyses, but nothing definitive. Another test is **high-energy cosmic rays**: could some ultra-high energy particles be debris from microscopic white hole eruptions? There were theories of Hawking radiation leaving imprints, but a direct link to white holes is unclear. OIPK specifically mentions **“white hole (upper vertex) ↔ black hole (lower vertex) connection”** – if taken literally, it sounds like it equates the entire beginning of our universe to a black hole from another realm. This is analogous to the proposal “our universe is inside a black hole.” Some papers have explored this mathematically: e.g., a Schwarzschild black hole metric’s interior can be continued to a new expanding universe domain (a different coordinate patch). So in principle, a black hole in a parent universe could contain a baby universe (with perhaps different physical constants in Smolin’s idea). It’s hard to find direct evidence for this from *inside* our universe, but maybe one could infer something like: if our Big Bang inherited certain conditions (rotation, charge?) from a black hole, we might detect that (e.g., if the whole universe had a slight rotation (Gödel-like or from a Kerr black hole source), or a slight asymmetry that could be relic of a black hole initial conditions). So far, we see no significant global rotation or strange asymmetry in our universe’s large-scale structure – it looks very uniform and non-rotating (any global rotation must be < 10^{-9} rad/yr or so, per constraints). That suggests if our universe came from a black hole, that hole was non-rotating and uncharged basically – a simple Schwarzschild. Which, if it had a parent, might mean the parent universe’s conditions were tuned such that a huge star collapsed without angular momentum (which is unusual). Nonetheless, the **information paradox** might find a resolution in OIPK’s framework through redundancy (the 12 mirrors) and connectivity (white hole outflows). If information can leak from black holes at the end of time or via alternate channels, then no firewall is needed and unitarity holds. OIPK’s fractal mirroring might mean that Hawking radiation is not the only escape – maybe information is mirrored into baby universes. This is not falsifiable at present, but one might consider indirect evidence: e.g., Penrose suggested anomalous low-entropy spots in the CMB could be evidence of Hawking radiation from a prior aeon’s black holes popping through [58]. Similarly, if a black hole in our universe spawned a baby universe, could we ever see any trace? Perhaps not, except maybe gravitational effects if that process requires a sudden mass loss from our side (but that likely happens only at the very end, beyond reach). Another possible empirical angle: **quantum gravity in the lab.** Some have considered that quantum black holes or analogues might be created in particle colliders if extra dimensions exist (so far none seen at LHC). If a microscopic black hole were created and then turned into a white hole (exploding), we’d detect an unexpected burst of particles. The absence of any such events at high energies doesn’t give much yet (as we haven’t produced any black holes at all).

Comparison to Other Frameworks: It’s helpful to situate OIPK relative to existing theories:

- **Causal Dynamical Triangulations (CDT):** Similarities: both have an explicit time foliation (slicing of spacetime) [9]. CDT uses discrete 2D/3D simplicial slices much like OIPK’s idea of 2D slices. Differences: CDT does not involve 12-directional mirroring or separate horizontal vs vertical time – CDT is still one time dimension, and space within a slice is treated conventionally (3D). Also, CDT’s slices are not “holographic screens” but rather spatial geometries that evolve. Compatibility: Possibly, if OIPK is discrete at Planck scale, one could imagine implementing it in a CDT-like model with each slice being a 2D lattice and connections ensuring the cross-relations (like identifying edges to mirror points). CDT already has lattice artifacts (e.g., a preferred time does not ruin diffeomorphism invariance in the continuum limit) [10], so having a structured lattice of symmetry might be okay as long as it’s ultimately summed over.

- **Wheeler–DeWitt (WdW) Equation (canonical quantum gravity):** Similarity: WdW has “no external time” – the wavefunction of the universe Ψ is defined on configuration space and obeys $H\Psi=0$. OIPK’s vertical implosion parameter could be seen as an internal time for a WdW equation (like the volume of universe). Difference: OIPK introduces effectively an extra time dimension, whereas WdW has timelessness at fundamental level. To reconcile, one could treat OIPK's τ as an internal time in WdW (say, $\Psi(x,y,z,\tau)$ with τ monotonic where z is like a clock variable). This is not standard WdW but an "unfreezing" by picking a clock. Barbour's timeless view contrasts with OIPK's dynamic collapse – though Barbour also has an "Janus point" which echoes OIPK's mirror. Compatibility: If OIPK is correct, the WdW equation might still hold but with boundary conditions at the mirror. It's an open problem in quantum gravity how to include multiple time flows, so OIPK might require going beyond WdW formalism (maybe a multi-time Schrödinger equation?).

- **Barbour's Platonia / Janus Point:** Similarity: time is an emergent sequence of "Nows," not fundamental; a special point (minimum complexity) is the divide for two time directions. OIPK also picks a special point (maximum expansion) as a divide. Differences: Barbour's model (with collaborators) typically has time-symmetry with a central point; OIPK implies time might not exactly reverse but at least an illusion of reversal or boundary. Barbour doesn't involve light or holography. Compatibility: They are philosophically compatible if one imagines shape-space coordinates describing each slice (Barbour's "time" is basically change of spatial relationships, which OIPK's vertical descent does). If τ is like Barbour's time parameter, then horizontal t might be something like a path parameter within a slice (Barbour doesn't have that though). Potentially, one could incorporate OIPK's framework into Barbour's by saying the configuration of the universe at a given instant is 2D (which is odd for shape space since we usually think 3D relations), but perhaps 2D holographic data determines 3D relations.

- **Holographic Principle ('t Hooft/Susskind):** Similarity: OIPK uses the idea that lower-D information can encode higher-D physics, with 2D slices encoding 3D reality, analogous to holographic screens. Also, the concept of entropy per slice aligning with $A/4$ is directly holographic. Differences: Standard holography uses a fixed boundary (like spatial infinity or a black hole horizon), not a stack of internal slices. Also, holography normally requires a specific spacetime with a certain symmetry (like AdS). OIPK's 12-fold fractal doesn't map to known dualities. Compatibility: Possibly a generalization – Erik Verlinde's entropic gravity allows many screens inside space [66]. One might conceive a *holographic bulk reconstruction* from a foliation of screens. However, one must be cautious: usual holography ensures *one* copy of information at the boundary, whereas OIPK's multiple slices might imply redundancy or even an over-counting if not constrained. The framework might need a rule like Bousso's: each light-sheet is a valid encoding, but one cannot freely use all at once beyond certain limits (to avoid violating the entropy bound). If OIPK satisfied Bousso's covariant entropy bound at all timeslices, it could be consistent with holography.

- **Verlinde's Entropic Gravity:** Similarity: Emergent gravity from entropy and information on holographic screens, notion of *entropy gradients driving motion*. If each 2D slice has an entropy distribution, the horizontal flow of information (light) might correspond to the diffusion of entropy that generates forces. Differences: Verlinde's model doesn't have multiple nested screens, he typically chooses a single spherical screen at radius enclosing mass [26]. OIPK effectively introduces *many* screens (each slice) and a fractal replication. Compatibility: Potentially, if one extended entropic gravity to a multi-layer system. Verlinde also recently proposed emergent gravity as an explanation for dark matter phenomena, based on entropy associated with cosmic information. If OIPK provided a natural place to store that entropy (like in the implosion axis or mirror images), it could give a physical realization to some of Verlinde's ideas. It's speculative, but maybe the "dark information" is in those unseen mirrored slices.

- **Loop Quantum Gravity (LQG):** Similarity: LQG has discrete spatial geometry (spin networks) and potentially discrete time steps (spin foams). If one truncated LQG states to something like an octahedral graph repeating, it might mirror OIPK's lattice. Differences: LQG doesn't single out a cubic lattice; it usually sums over all graphs (though one can choose a fixed graph for symmetry). Also, no explicit 12-fold fractal in standard LQG – though certain spin network states can be highly symmetric (e.g., a 6-valent node shaped like a cube as mentioned, or an intertwiner with octahedral symmetry). Compatibility: LQG could accommodate a state that has OIPK's symmetry as a *particular semiclassical state*. Perhaps a weave state that approximates an octahedral lattice of loops might look like OIPK microscopically. However, LQG by itself doesn't incorporate consciousness or two times. It does allow black hole to white hole transitions (Planck star results came from LQG) [67] [68], which is consonant with OIPK's BH–WH idea. But LQG would consider that happening for each black hole, not necessarily one for the whole cosmos (unless we treat cosmos as one black hole in some larger space).

- **String Theory:** Similarity: String theory has extra dimensions and rich symmetry (E8, etc.), and sometimes dualities that mirror theories (AdS/CFT being a 1-1 mapping of bulk to boundary info). Perhaps tenuously, the 12 directions might be related to a higher-dimensional lattice (E8 lattice in 8 dims can project to a 3D quasicrystal with icosahedral symmetry – icosahedral has 6 fivefold axes etc., not directly 12 edges, but E8 does involve golden ratio which appears in quasicrystals). Differences: Nothing in mainstream string suggests a cube-on-vertex spacetime or two times (except F-theory with 2 times, which is somewhat fringe within string). Also, string theory normally preserves Lorentz symmetry in 4D exactly (except in some non-critical or Lorentz-violating scenarios), so it wouldn't naturally include a fixed implosion frame. Compatibility: Possibly none – OIPK seems largely orthogonal to string theory paradigms. One could imagine maybe some exotic compactification yields an effective 12-fold symmetry in some potential, but that's speculation. Unless OIPK is formulated in a way that it emerges as an approximate description of some stringy state (for instance, maybe some brane collision scenario producing a cube grid of intersections?), it's not obviously connected.

- **Conformal Cyclic Cosmology (CCC):** Similarity: as discussed, CCC features a mirror-like identification of a final state with an initial state [69]. It is time-symmetric in a certain conformal sense and has the idea of a universe (aeon) preceding ours. OIPK also posits some form of time identification at expansion max. Differences: CCC's "mirror" is at infinite time in each aeon (requiring complete decay of matter to radiation), whereas OIPK's mirror is at finite time (peak expansion). Also CCC doesn't have repeats within an aeon, only at boundaries – OIPK might be single-bounce or fractal within one universe. Compatibility: If the maximum expansion is effectively like a crossover to a new aeon, OIPK could be a version of CCC where the cosmological constant is just enough to reach a static moment then recollapse (Penrose's original CCC expects no recollapse, just stretching out). If OIPK insisted on a recollapse (like an "anti-CCC"), one could contrive a scenario: imagine two aeons, one ours expanding, one time-reversed contracting, that meet at a common interface (the turnaround). That would be more like Tolman's oscillatory universe with a CPT identification at the crunch. Actually, an interesting new model by Turok and Boyle in 2018 *does* propose a CPT-symmetric universe where the Big Bang was a sort of mirror that created a twin universe on the other side of t=0 (with opposite arrow of time and CP-parities) [70] [71]. That model yields a natural explanation for why we observe an excess of matter over antimatter (the mirror universe gets the opposite). OIPK's mirror at maximum expansion rather than at the bang is like a shifted version of that – a mirror in the future instead of at the past singularity. Conceptually, if one had a symmetric bounce, one could have two arrows emanating from the bounce. So OIPK may well align with such CPT-symmetric bounce ideas, but places the bounce at the end of expansion rather than the beginning. In a closed universe, these are the same event if you consider one cycle only (Big Bang and Big Crunch are distinct though; CPT bounce model uses the Bang as the bounce, not a separate crunch). Possibly OIPK is envisioning the Big Bang and Big Crunch as a single White/Black hole pair connected (like one wormhole connecting them). In any event, CCC and CPT-symmetric cosmos are **similar frameworks** addressing time's arrow, so OIPK has some support from the fact serious physicists consider time-symmetric cosmologies plausible and even testable (e.g. via gravitational wave backgrounds or other subtle effects from previous cycles).

The above comparisons are summarized in the table below:

Framework	Similarities with OIPK	Key Differences	Compatibility
Causal Dynamical Triangulation (CDT)	Discrete time foliation (stacked slices) [9] ; a preferred global time coordinate.	No fractal 12-mirror structure; one time dimension only (no separate horizontal vs vertical times).	Generally compatible at a structural level – OIPK could potentially be implemented as a constrained CDT with symmetric lattice. No built-in consciousness aspect in CDT.
Wheeler–DeWitt Equation	Treats time as emergent (no external time); in OIPK vertical time could serve as internal time parameter.	OIPK has an explicit dynamic (implosion) whereas WdW is timeless until a clock is chosen. WdW doesn't allow two independent time flows.	Needs reconciliation – OIPK might be realized by choosing a time gauge in WdW (e.g., τ as time). Not naturally accounted for in WdW without extra structure.
Barbour's Platonia & Janus Point	Time as an ordered sequence of "Nows"; possible symmetric time reversal at a special point (Janus) [36] . Emphasizes timeless laws with time as emergent from change.	Barbour's time-symmetry is at minimum complexity, whereas OIPK's mirror is at maximum expansion (which could correspond to maximum volume, not minimum complexity). Barbour doesn't include any fractal mirroring or light-driven consciousness.	Philosophically compatible: OIPK's vertical implosion could be the progression through configuration space. If maximum expansion = minimum complexity, then OIPK's mirror could act like a Janus point. Requires assuming low entropy at both bang and turnaround, which is speculative.

Framework	Similarities with OIPK	Key Differences	Compatibility
Holographic Principle (general)	Information content of 3D volume given by 2D surfaces (slices as "screens") [24] ; entropy per area law suggests each 2D slice holds $S \sim A/4$ bits. Idea that our 3D experience is projection of 2D data.	Usually only *boundary* surfaces (at infinity or horizons) are used, not a stack of internal slices. Standard holography is one-to-one mapping, whereas OIPK has potentially many overlapping encodings (one per slice), raising questions of redundancy.	**Generalization needed:** Could be made compatible via Bousso's covariant holography, where each slice has a lightsheet and must obey entropy bounds [23] [24] . OIPK might be seen as a realization of holography in a cosmological setting (some researchers have tried "volume holography" in cosmology but it's unresolved). Not inconsistent, but unusual.
Verlinde's Entropic Gravity	Emergent gravity from information distribution on surfaces; entropic force from entropy gradients [26] [27] . OIPK's slices and horizontal flow could create such gradients (e.g., implosion might increase entropy on a slice, driving information outward).	Verlinde uses single holographic screen around masses, not fractal recursion. Also doesn't consider multiple time dimensions or cosmic bounces.	Potentially compatible if extended: OIPK could provide a physical picture for how information is distributed in space and time to cause gravity. For instance, as implosion reduces volume, information has to redistribute via horizontal light, possibly producing entropic forces. This is speculative, but not in direct conflict.
Loop Quantum Gravity (LQG)	Discrete spatial structure at Planck scale; possible resolution of singularities (BH to WH transitions via "Planck stars") [64] . LQG spin networks can have symmetric configurations (e.g., 6-valent nodes ~ cube) that might relate to OIPK's geometry.	LQG does not assume a preferred lattice like a cube; basis states are all possible graphs. Time in LQG is still single-parameter (though in spin foam it's discrete "ticks"). Consciousness not included.	Possibly OIPK could be a particular sector of LQG states (ones that form an octahedral lattice). LQC (Loop Quantum Cosmology) yields a bounce for the whole universe, akin to OIPK's bounce, but usually symmetric (no fractal 12 mirrors). LQG's Planck star idea supports BH→WH, aligning with OIPK on that point [64] . Overall compatible on quantum gravity aspects, but LQG doesn't address fractal structures or consciousness by itself.

Framework	Similarities with OIPK	Key Differences	Compatibility
String Theory (incl. F-theory)	Allows extra dimensions and even multiple time dim (F-theory has 2 times) [2]. Rich symmetry groups (E8 etc.) which in some cases relate to quasicrystal lattices in lower dimensions. String theory handles high entropy (e.g., via holography for black holes).	No known role for 12 discrete directions or fractal spacetime; string spacetimes are smooth (except at singularities) and usually Lorentz invariant (breaking that is beyond standard string theory). Also no explicit place for consciousness.	Largely *unclear:* OIPK doesn't map cleanly onto string frameworks. Perhaps a very exotic compactification or non-equilibrium string cosmology could mimic aspects (like a Milne universe piece matching a white hole). At present, string theory doesn't provide obvious support or refutation of OIPK, it just operates in a different paradigm (continuous, supersymmetric, etc.).
Conformal Cyclic Cosmology (CCC)	Time mirror at the end of the universe – future infinity of one aeon becomes Big Bang of next [69]. Implies a form of sequential universes and possible observable imprints from previous cycle [58]. Addresses the low-entropy puzzle by effectively reusing the final state as initial state (after conformal stretching) [72] [56].	CCC's "mirror" is at conformal infinity, requiring all mass to decay (very far future). OIPK's mirror is at a finite time (turnaround) – a different mechanism. Also CCC does not incorporate fractal sym or multiple time flows; it's classical until the crossover which is geometrical.	Possibly **analogous:** If OIPK's maximum expansion is treated like the interface to a new aeon, then OIPK could be a variation of CCC with a bounce. The presence of 12-symmetry and consciousness are extra features CCC doesn't have. Penrose's idea of information passing to next aeon via Hawking radiation [56] is conceptually similar to OIPK's notion that info isn't lost (just mirrored). Both share a view that Big Bang was a special, low-entropy "mirror" state.

Framework	Similarities with OIPK	Key Differences	Compatibility
CPT-Symmetric Universe (e.g., Boyle–Turok)	Universe is time-symmetric around the Big Bang (our universe has a CPT-reflected mirror universe extending backward in time before t=0) [70] . Solves certain problems like baryon asymmetry by mirror accounting. Time's arrow emerges from that symmetric start.	That model's mirror is at the Big Bang, not at maximum expansion. It produces *two* universes diverging in opposite time directions from the bang, whereas OIPK might produce two converging at the crunch or one bounce turning single trajectory.	If OIPK's maximal expansion is analogous to t=0 in CPT model, then after that point the universe effectively goes CPT-reversed. This would mean our future is a mirror universe contracting while time runs reversed from our perspective. It's a crazy picture but mathematically akin to gluing two CPT halves at turnaround. Not mainstream, but *conceptually* within realm of speculative cosmology.

(Table: Comparison of OIPK with various theoretical frameworks, highlighting overlaps and divergences.)

Section 1: Mathematical Foundations

1.1 Differential Geometry of the "Cube-on-Vertex" Spacetime: To formalize OIPK, we consider a spacetime manifold structured by a foliation into 2D surfaces. Let's denote coordinates (x, y) for the 2D slice (horizontal plane), and a coordinate z for the vertical direction (the axis of implosion). Additionally, we maintain a time coordinate t for physical processes on each slice (horizontal time). The metric ansatz suggested is:

$$ds^2 = -c^2\,dt^2 + a(\tau)^2\left(dx^2 + dy^2\right) + b(\tau)^2\,dz^2,$$

where τ is some function of z (or the proper time along z-worldlines). Here $a(\tau)$ represents the scale factor for the 2D slices and $b(\tau)$ the "metric" in the vertical direction. This form is reminiscent of an **anisotropic cosmology** – specifically, a (2+1)+1 split (two spatial dimensions treated one way, the third spatial dimension differently). It's akin to a **Bianchi I universe** but with two equal scale factors (for x,y) and one different (for z). If $b(\tau)$ shrinks while $a(\tau)$ grows, that corresponds to an implosion along z combined with expansion in the x–y plane (or vice versa for contraction phase). Such metrics have been studied: e.g., a **Kasner solution** in a 3+1 universe with two exponents equal and one different. Kasner's solution (vacuum GR in a closed universe) has $a(t) \propto t^p$, $b(t) \propto t^q$ with $2p+q=1$ and $2p^2+q^2=1$ for exponents p,q. One solution is $p=2/3, q=-1/3$ for example, which means two directions expand and one contracts (this is a possible qualitative analog to OIPK's behavior inside a black hole or in an anisotropic cosmology). To incorporate a bounce, one can use a sinusoidal or piecewise scale factor: OIPK's prompt suggested:

- **Sinusoidal form:** $a(\tau) = a_{\min} + (a_{\max}-a_{\min}) \sin\left(\frac{\pi \tau}{\tau_{\text{total}}}\right)$, which increases then decreases. This would naturally reach $a_{\max}$ at $\tau = \tau_{\text{total}}/2$ (halfway) – representing maximum expansion. $b(\tau)$ might inversely mirror that if volume is conserved or something, or possibly $b(\tau)$ could be monotonic if the total 4-volume is something specific. Alternatively, a piecewise linear expansion-contraction was given, but that's not smooth. The sinusoidal or an analytic function would be better for field equations.

Given such a metric, one can in principle compute the Einstein field equations. The presence of two scale factors means there will be an anisotropy stress (unless some matter fields create that difference). Typically, for anisotropic universes, one might include a **magnetic field** or an **axion field** to induce differences in expansion rates, or simply accept anisotropy in vacuum as in Kasner. If OIPK's fractal symmetry has a stress-energy manifestation, it could come from something like a network of domain walls or a lattice of cosmic strings aligned to form a web (but a 2D web in 3D space – not common). Possibly, one could treat the 12 mirror directions as akin to 12 families of non-intersecting straight cosmic strings passing through every point (that's just brainstorming) – but 12 independent directions of stress would likely produce an isotropic average (if fully symmetric). To get a stable cube symmetry, you might need an isotropic stress (like radiation) plus some higher curvature or solid-state feature (like a "spacetime crystal" concept).

A simpler approach: consider $T_{\mu\nu}$ that respects the symmetry of the metric ansatz. The isometry group of a cube-on-vertex (an octahedral group) is discrete, but continuous symmetries include translations along x,y and probably an $O(2)$ rotation in the x–y plane if we assume the slice is isotropic. There is no continuous symmetry involving z except reflection $z\to -z$ perhaps if top-bottom symmetrical (if the diamond is symmetric about midplane). For solving Einstein's equations, one might assume a perfect fluid form $T^\mu_{\ \nu} = \mathrm{diag}(-\rho, p_{x}, p_{y}, p_{z})$ with $p_x = p_y \neq p_z$ (two equal pressures and one distinct – this is known as an **anisotropic fluid**). Then the Einstein equations reduce to two Friedman-like equations: one for the x–y directions and one for the z direction. Without delving into algebra: they will impose conditions like $\frac{\ddot{a}}{a} = -\frac{4\pi G}{c^2}\left(\rho + \frac{p_x + p_y}{2}\right)$ and $\frac{\ddot{b}}{b} = -\frac{4\pi G}{c^2}\left(\rho + p_z\right)$ (schematically), and a constraint $\frac{\dot{a}^2}{a^2} + 2\frac{\dot{a}\dot{b}}{ab} = \frac{8\pi G}{3c^2}\rho - \frac{K}{a^2}$ if K is some curvature (maybe 0 if spatially flat in x,y). For a bounce, we'd need $\dot{a}=0$ at max expansion but $\rho>0$, which typically requires negative effective pressure (like a cosmological constant or scalar field) to turn expansion into contraction. That implies something like p_z or p_{xy} must be sufficiently negative around the turnaround. Perhaps the **scalar field** (inflaton-like) that drove expansion could do a slow-down and reversal.

1.2 Orthogonal Temporal Foliation: If one insisted on treating t and τ both as time coordinates in a 5D sense, one could consider a metric with signature (2,3) to include both. But physical interpretation is tough. Instead, it's better to think of τ as a parameter labeling slices. The OIPK prompt even suggests an evolution equation $\partial^2 \Psi / \partial t \partial \tau = ...$ – a cross-term coupling in a wave equation. A toy wave equation in such background might be:

$$\frac{\partial^2 \Psi}{\partial \tau \partial t} = v(\tau)\nabla^2_{xy}\Psi + u(\tau)\frac{\partial^2 \Psi}{\partial z^2},$$

just guessing $v(\tau)$ etc. But actually, since z coordinate is like space, maybe they meant an equation mixing the two "times." Perhaps they envision a master equation where τ-derivative (vertical evolution) is driven by some operator that includes t-derivative. In two-time physics (Bars), one usually finds a constraint that relates dynamics in the two times such that effectively one time emerges for observed quantities.

It might be easier to consider a Hamiltonian formalism: we have a Hamiltonian constraint (like WdW) and perhaps a momentum constraint, and one could gauge-fix one degree of freedom as "vertical time" and another as "horizontal time." The interplay could yield an equation akin to $\frac{\partial}{\partial

\tau}\left(\frac{\partial \Psi}{\partial t}\right) = \text{(some operator)}\Psi$. Without a concrete Lagrangian it’s speculation.

1.3 Recursive Mirroring as Group Action: Mathematically, the 12 mirror transformations can be represented as elements of the **octahedral group O_h** (or its rotation subgroup O). The 12 edges of a cube correspond to 180° rotations about axes through opposite edges of the cube (these are order-2 elements in the rotation group). There are exactly 6 such axes (each axis goes through two opposite edges), and rotating 180° about each yields a distinct symmetry. That’s 6 transformations (these combined with identity might form a $\mathbb{Z}*2^6$ subset, but they’re not independent as a group since combining two gives others). If we include reflections, we can reflect across planes containing those axes or others. The result is a non-abelian group of 24 rotations (or 48 with reflections). A recursive mirroring means we consider the images of a point under repeated action of these group elements. Provided the group is finite (24 elements), repeated application just yields those finite images (no fractal – just an orbit of up to 24 points). However, if one considers not strictly symmetric mirror planes but slightly offset (like kaleidoscope with angle not exactly dividing 180), one can generate infinite patterns – sometimes fractal if inside a sphere (like Schwarzschild tiling on hyperbolic space can yield fractals on the boundary). Perhaps OIPK implies a fractal in the sense of self-similar across scales rather than literally infinite distinct orientations. If at each scale of structure, the same 12 directions appear (maybe rotated or permuted), then one gets a fractal. This is like saying the Coxeter group is applied at different length scales (a dilation symmetry in addition to rotation/reflection sym). For instance, consider an Iterated Function System (IFS): define 12 contraction maps (each maybe a rotation about an edge direction and a scaling by factor <1). The fixed set of such an IFS could be a fractal of octahedral shape. The fractal dimension D solves $ \sum*\approx0.289$, then $D=2$ (space-filling threshold). Achieving fractal dimensions between 2 and 3 could match cosmic structures which have fractal dimension ~2.x on certain scales.}^{12} r_i^D = 1$ where r_i are the scale factors. If all equal r, then $12 r^D = 1$, so $D = \frac{\ln(1/12)}{\ln r}$. If r is, say, 1/2, then $D \approx 3.585$ as earlier estimated. If r smaller, D bigger (closer to 4 if r small). If $r=1/\sqrt{12

1.4 Gauge Group Mapping: The mention of gauge groups like $E8$ or $SU(12)$: $SU(12)$ is 143-dimensional group, probably a red herring. $E8$ is exceptional with 248 dimensions, known to have symmetry connections to quasicrystals (the $E8$ root lattice projects to 8D and 4D structures with icosahedral symmetry, not cubic). Possibly the user wonders if 12 mirror sym might tie to a group like A_4 (the symmetry of a cube’s diagonal – but A_4 is 12-element group isomorphic to rotational symmetry of a tetrahedron, not cube edges). Actually, the rotational symmetry group of an **icosahedron** is order 60, which includes 6 fivefold axes, 10 threefold axes, 15 twofold axes – not directly 12 of one type. The **dihedral group D6** has 12 elements (rotation/reflection sym of a hexagon). But cube edges maybe correspond to the idea of 6 orthogonal mirror pairs, so maybe $\mathbb{Z}_2^3$ semi-direct something.

This is esoteric, but one might find that the *binary icosahedral group* (double cover of rotational icosa group) is $SL(2,5)$ or something with order 120 – again no obvious 12. $SU(2)$ had discrete subgroups including one of order 24 (binary tetrahedral, corresponds to 24-cell?).

If one is bold, one could imagine each mirror direction introduces a kind of discrete transformation akin to a quaternion operator (like i,j,k axes). The 12 edge directions could be seen as ±i, ±j, ±k (6) plus ±(i+j+k), ±(i+j-k), etc (not exactly edges – edges of cube if oriented differently). Perhaps there's some algebra where 12 such reflections generate a known polyhedral group.

1.5 Causal Structure: The vertical implosion suggests a *preferred frame*. Locally, presumably, physics is Lorentzian, so one must define how an event on one slice connects to an event on the next. If slices are separated by timelike intervals (vertical direction is timelike), then dz path is timelike (with metric

$b(\tau)^2 dz^2$ possibly >0 if b is imaginary or metric sign flip). If $b(\tau)$ were pure imaginary, that'd make dz timelike, but that's not typical. Instead, we had dt timelike. So likely z is spatial (the vertical axis is spatial coordinate albeit with contracting scale factor). That means the universe is actually 3D space with unusual metric anisotropy. So then what is "implosion"? It's not time itself but space shrinking – in effect, it's a collapse of 3D volume if b decreases. But one still has only one time (t). So maybe OIPK doesn't literally add a time coordinate but uses an anisotropic cosmology with a special spatial dimension that shrinks.

However, they described vertical implosion as if it's "temporal": perhaps they imagine falling inward in a higher-dimensional space? Another interpretation: treat the 4D spacetime as embedded in higher dimension so that the "vertical" is an extra spatial dimension pointing towards a collapse point (like the way an FRW universe can be embedded in a higher-d space as a hyper-surface). If so, then vertical "implosion" could be the extrinsic curvature effect. This is speculative, but one could try to embed a closed FRW in 5D Minkowski: the FRW expansion becomes motion in the fifth dimension. If OIPK can be realized that way, the vertical coordinate might be such an embedding coordinate. But they phrase it as if inside the 4D world.

Given ambiguity, we might consider two interpretations:

(A) *2+1+1 decomposition:* The universe is 4D with metric as above (t is time, one special space axis z). Then "implosion" just means the z-direction length scale shrinks (like a pancake collapse). At some point, $b(\tau) \to 0$ hitting a singularity – possibly a big crunch (or at past end a big bang). Meanwhile $a(\tau)$ might start from 0 at Big Bang (point), expand to maximum, then recollapse? Actually if at Big Bang $a=0,b=0$ (point), it expands a outward and b grows (?), or maybe b always shrinks, not likely – maybe b grows then shrinks for a closed cosmic cycle.

(B) *Two time coordinates (t and τ) in 4D param-space:* Then effectively the metric has Lorentz signature (2,2) or something. But physically we only experience one time. So probably not.

We lean with (A) which is just an anisotropic closed universe (like mix of Kantowski-Sachs and closed FLRW). If we consider the entire spacetime as a "diamond," the top point could be a **white hole singularity** (like a Big Bang point in a closed model), the bottom a **black hole singularity** (Big Crunch). The maximal cross-section in middle is when a peaked and b maybe symmetric.

This indeed resembles the Schwarzschild interior metric: inside a Schwarzschild black hole (for $r<2M$), the t and r coordinates swap roles – r becomes timelike (decreasing to 0 at singularity). That interior metric can be seen as an anisotropic cosmology: one space coordinate (angles on 2-sphere) and one timelike radial coordinate. In fact, the Schwarzschild interior is like a Kantowski–Sachs universe: metric $-d\tau^2 + (R/\tau)^2 dz^2 + \tau^2 d\Omega^2$ (just symbolically), which is qualitatively similar to OIPK metric if x,y were angular coordinates of a 2-sphere and z radial proper distance. So consider: maybe the OIPK universe is *the interior of a massive black hole.* The upper white hole vertex is where the hole formed (or a previous universe collapsed into it, which appears to us as a Big Bang-like white hole event). The lower black hole vertex is the future singularity we'll reach. That would unify the BH–WH connection and the cosmology in one: We live **inside** a black hole that formed from some parent universe collapse – our Big Bang was the white hole event of that collapse. This scenario has been hypothesized by others. It naturally gives a closed universe (since black hole interior can be spatially closed in some models), and it means at "maximum expansion" we might be at the bounce point before recollapse to singularity. In Schwarzschild interior, the spatial 2-sphere radius first grows from 0 at the white hole singularity (the Big Bang inside) to a maximum (at the horizon radius) then decreases to 0 at the black hole singularity. That horizon radius would be our maximum $a(\tau)$ (though in standard black hole interior it's constant in time but coordinate wise one can conceive an apparent horizon size that changes if embedded in larger cosmos). If our entire universe is within a black hole of another universe, some puzzles are answered (why our Big Bang might have low entropy – it's essentially a state arising from collapse).

No direct evidence for that, but mathematically it's appealing. If true, the **12 mirrored directions** could be something to do with degrees of freedom of gravitational collapse (maybe anisotropic collapse yields multiple "jets"? Or quantum gravity yields multiple branches). In any case, this is a potential formal approach: treat OIPK as a Kantowski–Sachs metric (for BH interior type) or a Bianchi IX (mixmaster) which can have chaotic reflections (mixmaster anisotropy famously is described as billiards in hyperbolic space reflecting off walls corresponding to different anisotropy potentials – interestingly **mixmaster cosmology** has an infinite series of "reflections" between anisotropic axes as one approaches singularity, which is mathematically related to the Weyl group of E_{10} or E_8 in certain formulations!). This is a fascinating parallel: the BKL (Belinskii–Khalatnikov–Lifshitz) model of a chaotic Mixmaster universe describes the evolution as a sequence of Kasner epochs where the axes of contraction "bounce" between the three spatial axes due to curvature effects – effectively the contraction sloshes between x, y, z directions. The dynamics is often visualized as a point moving in a triangular billiard in hyperbolic space, reflecting off walls (representing when one scale stops contracting and another starts). The pattern of reflections is fractal and related to an infinite Coxeter group. In some sense, that is **"recursive mirroring along different directions."** Possibly OIPK's 12 edges hint at some truncation of mixmaster (maybe if the billiard were an ideal triangle, the system would have a certain symmetry). Indeed, mixmaster's symmetry is tied to the group $SL(2,\mathbb{Z})$ or the Weyl group of E_10 etc. But maybe a simplified model yields 12 distinct orientations and then repeats – who knows. If we could tie OIPK fractal to mixmaster chaos, that would be a strong theoretical angle, as mixmaster oscillations are a well-studied feature of GR near singularities.

1.6 Summary of Math Tools: In validating OIPK mathematically, one should:

- Solve Einstein's equations for the anisotropic metric to see conditions for a bounce and what matter (or quantum effect) needed.
- Investigate whether repeated "reflections" can arise from anisotropic chaos (BKL theory) – that might connect the 12 mirror metaphor to known chaotic oscillatory behavior near singularities.
- Possibly connect to **Regge calculus**: think of a spacetime triangulated by diamond-shaped 4-simplices that align in a hypercubic way. "Cube-on-vertex" suggests perhaps using an **octahedral lattice** (since the Voronoi of a cubic lattice is a rhombic dodecahedron, which might tile space; and dual to that is a body-centered cubic lattice tiling with octahedra…). Causal Set or Dynamical triangulations on an octahedral lattice could be attempted – though usually CDT uses simple uniform triangulations.
- Consider if the vertical direction could be an intrinsic time coordinate (like York's extrinsic time or something) while horizontal is the usual time – but that's double counting.

In conclusion, mathematically OIPK can be cast as a specific anisotropic cosmology (most likely a closed Kasner-like universe) possibly corresponding to an embedded black hole interior. The **cube symmetry** is unusual but could be enforced by initial conditions or a potential that favors that mode (imagine a universe filled with a network of domain walls forming a cube lattice – that could impose a cubic anisotropy). The **field equations** would require anisotropic stress or higher-curvature terms to allow a bounce (since ordinary matter with positive density can't bounce a closed universe without exotic ingredient like scalar field or negative pressure). Perhaps the 12-direction symmetry hints at a link with **dodecagonal or polyhedral configurations** in potential (some cosmological potentials have hidden symmetries).

One more note: **Causal structure** – since t is the only timelike coordinate, light cones are oriented mostly along t, and as long as $b(\tau)$ remains positive, z is spacelike. If $b(\tau)$ crosses 0, that's a singularity (like mixmaster has $a_i(t)\to0$ sequentially – each time one spatial scale goes to 0, one might interpret a temporary lightlike nature). But ignoring that, the model is globally hyperbolic (sliced by t).

We might mention briefly the concept of **multiple time dimensions** in physics to see if any formal metric used that. F-theory's (10,2) signature was noted [2] ; also certain ad hoc metrics like in "two-time physics" gauge constraints ensure no extra physical degrees of freedom even with two times [1] . But OIPK likely doesn't want to break too far from conventional physics; it just reinterprets one spatial dimension as doing something time-like (imploding).

We'll likely finalize Section 1 with the metric, mention mixmaster reflection idea, and highlight that a bounce requires exotic matter or quantum gravity (like loop quantum bounce or ekpyrotic scenario which can generate a turnaround with contraction to expansion).

Lastly, note the **White hole / black hole solution**: There is a known analytical extension called the **Kruskal–Szekeres diagram** that shows a white hole region (past singularity), our universe region, black hole interior, and another universe region. If one identifies our Big Bang as a white hole from a prior collapse, one essentially glues that region accordingly. Some studies have looked at matching a Friedmann expanding universe onto a white hole interior solution. It's tricky because FRW is homogeneous, white hole (Schwarzschild) is vacuum around – but maybe for a large black hole the interior can behave like a vacuum-dominated FRW (with heavy curvature).

Given space, perhaps it's enough to outline these ideas.

Now I will write Section 1 accordingly.## Section 1: Mathematical Foundations

1.1 Spacetime Metric and Foliation: We formalize OIPK's spacetime as a foliation of 2-dimensional spatial slices (the "upright" time-slices) indexed by a global parameter. Let coordinates (x,y) span each 2D slice, and let z denote the vertical direction (from upper vertex to lower vertex). Additionally, we maintain a time coordinate t for physical processes *within* each slice (the horizontal, light-propagation time). A suitable metric ansatz is:

$$**ds^2 = -\,c^2\,dt^2 + a(\tau)^2\,(dx^2 + dy^2) + b(\tau)^2\,dz^2,$$

where τ is a monotonic function of z (one may choose $\tau = z$ in appropriate units, or τ as the proper time along worldlines at (x,y) fixed). Here, $a(\tau)$ is the **scale factor** of the 2D slices and $b(\tau)$ is the **vertical scale factor**. This metric has the form of an **anisotropic cosmology** – specifically, a (2+1)+1 dimensional split. It assumes the 2D slices are isotropic in themselves (invariant under rotations in the xy-plane) but the z-direction generally evolves differently. Such metrics are well-studied in general relativity as special cases of Bianchi I (if slices are flat) or Kantowski–Sachs models (if slices are closed surfaces).

For intuition: if $a(\tau)$ grows while $b(\tau)$ shrinks, space expands in the xy-plane but contracts vertically – *a pancake-like cosmology*. If $b(\tau)$ goes to zero at some τ_{final}, that signals a collapse into a lower-dimensional state (a "crunch" of the vertical dimension). Conversely, if $a(\tau)$ vanishes at endpoints (like $\tau=0$ or τ_{final}), that signals a collapse of the horizontal dimensions (perhaps initial or final singularities). The OIPK narrative describes an initial expansion from a point (upper vertex) to a maximum spatial extent, then re-contraction to a point (lower vertex). We can accommodate this by specifying appropriate $a(\tau)$ and $b(\tau)$ profiles. One convenient choice is:

- **At $\tau=0$ (upper vertex = Big Bang/white hole):** $a(0)=0$ (the 2D slices start from a singular point) and possibly $b(0)=0$ (the vertical extent also starts at zero). This is a "point" origin – analogous to a Big Bang singularity, but one could allow $b(0)$ to be nonzero if the initial singularity is a line or extended in z; however, for symmetry we take it as a point.

- **For $0<\tau<\tau_{\max}$:** $a(\tau)$ increases, representing expansion of space, while $b(\tau)$ initially increases (if one imagines the white hole opening up) but later may decrease. At $\tau=\tau_{\max}$ (the mid-point of the diamond), we expect **maximal spatial volume.** In a closed cosmology, this is where expansion halts and contraction begins. So we impose $\dot{a}(\tau_{\max})=0$ (turnaround of horizontal expansion). We might also have $\dot{b}(\tau_{\max})=0$ if the vertical direction similarly turns around there, *or* it could be monotonic (depending on model – more on that below). If the universe is symmetric about the midpoint, one would have $a(\tau)$ symmetric and $b(\tau)$ symmetric about $\tau_{\max}$.

- **For $\tau_{\max}<\tau<\tau_{\text{final}}$:** $a(\tau)$ decreases toward 0 as $\tau \to \tau_{\text{final}}$ (Big Crunch), and $b(\tau)$ likely decreases to 0 as well (the vertical line pinches off). At $\tau=\tau_{\text{final}}$ (lower vertex), $a(\tau_{\text{final}})=0$, $b(\tau_{\text{final}})=0$. This is the final singularity (a black hole type singularity from inside view).

A simple choice that qualitatively fits this is to take $a(\tau)$ as **sinusoidal** and $b(\tau)$ as perhaps constant or also sinusoidal. For example:

$$a(\tau) = R\sin\Big(\frac{\pi\tau}{\tau_{\text{final}}}\Big), \qquad b(\tau) = B\sin^n\Big(\frac{\pi\tau}{\tau_{\text{final}}}\Big),$$

with constants R, B setting length scales and n some power. If $n=1$, $b(\tau)$ bounces similarly to $a(\tau)$ (vanishing at both ends and peaking at mid). If $n=0$, b is constant (the vertical size doesn't change – unlikely if we truly have an implosion). If $n=2$, for instance, b is more sharply peaked (it shrinks faster). The exact form would be determined by solving Einstein's equations for the given matter content (see below). Another possibility given in the prompt was piecewise linear: grow linearly to a max, then linear decrease – which is rough but captures the idea of an expansion phase then contraction phase.

1.2 Dynamics via Einstein's Equations: The above metric is of **Friedmann-like form with anisotropy**. To find $a(\tau)$ and $b(\tau)$, we must specify the stress-energy tensor $T_{\mu\nu}$. By symmetry of the 2D slices, the stress-energy should at least satisfy $T^x{}x = T^y{}_y$ (equal pressure in x and y directions, call it $p}$ for horizontal pressure) and possibly a different pressure in z direction $T^z{z = p_t = -\rho$ for a perfect fluid). So in component form:}$. We can let the energy density be $\rho = T^t{}_t$ (with sign convention $T^t{

$$T^\mu_\nu = \mathrm{diag}(-\rho(\tau),\ p_h(\tau),\ p_h(\tau),\ p_v(\tau)),$$

assuming homogeneous composition on each slice (no x,y dependence). This is an **anisotropic fluid** or "Bianchi I type" stress tensor. Many forms of matter can produce $p_h \neq p_v$. For example: - A uniform **magnetic field** oriented along the z-axis would produce pressure $p_h = (B^2/2\mu_0)$ in the xy plane and $p_v = - (B^2/2\mu_0)$ along z (because magnetic field lines resist compression transverse to them but not along them). This is one way to get anisotropic pressure. - A **cosmic string**

network lying in the xy plane would contribute tension in one direction but not others. - Or, more generally, a gradient of a scalar field could break isotropy.

For simplicity, one can first treat a *vacuum solution with anisotropic curvature* (this leads to the Kasner solutions) or include a cosmological constant to allow bouncing. The Einstein equations (in units $8\pi G = c = 1$ for brevity) for the metric above yield (dot means $d/d\tau$):

- **Friedmann constraint:**

$$\frac{\dot{a}^2}{a^2} + 2\,\frac{\dot{a}\,\dot{b}}{a\,b} + \frac{k}{a^2} = \frac{1}{3}\,\rho,$$

  where k is the 2D slice curvature (for a flat 2D plane $k=0$; if slices are closed like a 2-sphere, k would be $+1$ times something, but here slices are 2D infinite plane if not compactified. We might consider them closed 2D surfaces – e.g., a 2-sphere – but the metric form given is more like infinite plane. We could make x,y periodic to effectively have a 2-torus topology, which is topologically possible but a cube-on-vertex suggests maybe a closed surface like a topologically spherical or toroidal slice. However, 2-sphere doesn't embed in 3D Euclidean nicely; a cube-on-vertex might hint slices are finite and closed. For now assume $k=0$ for simplicity, meaning slices are flat or large enough that curvature is negligible during most evolution).

- **"Raychaudhuri" equations for scale factors:**

$$\frac{\ddot{a}}{a} + \frac{\ddot{b}}{b} + \frac{\dot{a}\,\dot{b}}{a\,b} = -\,\frac{1}{2}\,(\rho + p_h),$$

$$2\,\frac{\ddot{a}}{a} + \frac{\dot{a}^2}{a^2} = -\,(\rho + p_v).$$

  These come from $G^x{}_x = G^y{}_y$ and $G^z{}_z$ components of Einstein's equations.

In the special case of vacuum ($\rho = p_h = p_v = 0$), these reduce to the Kasner conditions. The Kasner solution is known to satisfy $\frac{\ddot{a}}{a}=0$ and $\frac{\ddot{b}}{b}=0$ in vacuum, leading to power-law behavior: $a(\tau) \propto \tau^p, b(\tau) \propto \tau^q$. Then the constraint demands $2p+q = 1$ and $2p^2 + q^2 = 1$ [73] [9] (for vacuum in 4D, these are the Kasner exponent conditions). A particular solution is $(p,q)=(\frac{2}{3}, -\frac{1}{3})$. That solution has $a(\tau)\propto \tau^{2/3}$ expanding and $b(\tau)\propto \tau^{-1/3}$ contracting – interestingly, it means the cross-sectional area $a^2 b = \tau^{2/3} \tau^{-1/3} = \tau^{1/3}$ still grows, but anisotropically. However, Kasner has a singularity at $\tau=0$ and no turnaround – it's monotonic in one direction of time (no bounce). To get a bounce, *exotic stress-energy* is needed: e.g., a positive cosmological constant or phantom field to cause $\ddot{a}>0$ at some point, or quantum gravity corrections.

One simple way to allow a bounce is to include a **cosmological constant Λ** (which has $\rho = \text{const}$, $p_h = p_v = -\rho$). A positive Λ tends to inflate all directions, whereas a negative Λ can cause recollapse. For a closed universe, a positive Λ could cause a bounce by preventing $\dot{a}$ from going negative. Conversely, a negative Λ could ensure recollapse. Because OIPK expects recollapse, we might consider a scenario of a closed 2D slice ($k>0$) plus negative Λ (which pulls back). If finely tuned, the universe could reach a maximum then crunch. Alternatively, a scalar field with appropriate potential can yield a bouncing solution (like in ekpyrotic or cyclic models, where a scalar field with negative potential energy can cause a contracting phase that transitions to expansion).

Key point: *The mere topology of a cube-on-vertex does not by itself solve the bounce – new physics is needed to avoid singularities.* Leading candidates: **Loop quantum cosmology** effects (which replace the $a=0$ singularity with a bounce by quantum gravity repulsion [64]), or a **CPT symmetric extension** (where time passes through the crunch smoothly by matching to an expanding solution in a time-reflected manner [70]). In absence of such, classical GR with normal matter would hit singularities at both ends (no bounce, just a start and end). OIPK seems to imply one connected history from white hole to black hole, presumably through quantum gravity bridging the endpoints (we will discuss that in Sec. 7 on BH–WH).

1.3 Orthogonal Temporal Coordinates: We stress that in the metric above, t is the only explicit time coordinate. The "vertical implosion" parameter τ plays the role of a label for slices (one might casually think of τ as a second time axis, but physically it's more like a **curvilinear spatial coordinate** or an internal time parameter for the cosmic evolution). One might be tempted to treat (t, τ) as two time coordinates. If we did consider an extended 5-dimensional manifold with coordinates (t, x, y, τ) and perhaps an extra dimension for closure, we could attempt to embed this 4D anisotropic metric in a higher dimension. For instance, some cosmological models can be embedded in a 5D space where the extra dimension's coordinate acts like an evolution parameter. However, within standard 4D general relativity, we have just one time dimension – OIPK does not truly add an extra time in the fundamental sense, it just splits the evolution in an "orthogonal" fashion (space evolves in one direction differently than in others).

This is a critical subtlety: The **orthogonality** of the dynamics refers to the separation of degrees of freedom rather than a second temporal axis that observers experience. Observers in OIPK have worldlines that progress in t (their clocks measure t). The vertical collapse τ is more of a global parameter like cosmic time in FRW – no individual observer "feels" τ separately; it's inferred from the changing conditions of the universe as a whole. One can think of τ as *cosmic time* and t as *local light-propagation time*, but physically these could be related (for instance, τ might coincide with t for comoving observers at some reference). In a homogeneous model, often one sets $t=\tau$ by choosing a gauge (proper time of comoving frame). But OIPK distinguishes them conceptually: horizontal light speed c is measured with respect to t, while the rate of collapse $v_{\text{imp}} = d b/dt$ could be different. If one defines $v_{\text{imp}} = \frac{db}{dt}$ and compares to c, indeed one can have $v_{\text{imp}} \neq c$ without contradiction, since $b(t)$ is not a signal velocity but a metric scale rate. There is no violation of relativity in having metric degrees shrink superluminally – e.g., during inflation the scale factor expands faster than light separation. So the "asynchrony" is just that: the metric's z-extent could diminish at a rate not equal to c. Provided no physical information is carried superluminally in a local sense, causality holds. In our metric, local light cones are determined by $ds^2=0$: $c^2 dt^2 = a^2 dx^2 + a^2 dy^2 + b^2 dz^2$. A photon moving purely vertical ($dx=dy=0$) satisfies $dz/dt = c\,/b(\tau)$ (which in general is $\neq c$ if $b\neq 1$). If $b(\tau)<1$ (in appropriate units), a vertical photon crosses the universe faster than horizontal light would cross an equivalent distance – but this "faster" is just because z distance is physically shorter by scale factor b. There is *not* an extra time dimension, just anisotropic scaling. Thus we conclude: **OIPK's two "times" (horizontal t and vertical parameter τ) can be interpreted within a single-time framework** – τ essentially tracks the time-evolution of the metric (like cosmic time), while t is the time coordinate for propagation within each slice. One can choose a gauge where $t=\tau$ (proper time slicing), but the distinction is conceptually useful to separate the two dynamical modes (photon motion vs implosion motion).

Nonetheless, to validate OIPK's consistency, one must ensure that introducing this anisotropy doesn't break fundamental symmetries too badly. Locally, Lorentz invariance is broken by the global anisotropy (there is a preferred spatial direction z). However, as long as this is on cosmological scales, local physics still *approximately* sees isotropy (especially if $b(\tau)$ evolves slowly or is near unity in the current epoch). Constraints on such anisotropy exist (from CMB isotropy, etc.): observationally, the universe is isotropic to one part in 10^5. So if today's a and b are not equal, we'd detect e.g. different expansion rates or redshifts in different directions. The OIPK framework might require that we are nearly at the symmetric turnaround (so that expansion in all directions is currently similar). It could also rely on the fractal mirror structure to somehow isotropize observations (a possibility if multiple mirror images mimic isotropy). But generally, a strong anisotropy in expansion is disfavored by data [74]. Thus, OIPK likely must have $a(\tau)$ and $b(\tau)$ nearly equal for much of the history since last scattering, or else provide a mechanism (like multiple domains oriented differently whose average appears isotropic). Some cosmological models (e.g., mixmaster universes) can isotropize due to inflation or certain matter domination periods. Perhaps OIPK had an early inflation that drove $a\approx b$ and only at approach to the crunch does b drastically shrink. If maximum expansion is near our era, b might just now be starting to significantly deviate, which could be connected to the observed acceleration (speculatively: if b starts decreasing, it could mimic a dark energy effect as the volume expansion in horizontal directions outpaces vertical). This is conjectural but illustrates how to reconcile with data: OIPK's exotic features might only become pronounced as we near the turnaround, which might be soon in cosmological terms (e.g., perhaps a few times the current age). We discuss possible signs (like Hubble tension or axis alignments) in later sections.

1.4 Recursive Mirroring and Group Theory: The **"12-directional recursive mirroring"** suggests a discrete symmetry. A cube has 12 edges, and placing a cube on a vertex (diamond orientation) makes those edges radiate in a symmetric pattern. What mathematical object has 12 directions of symmetry? The **octahedron** (dual of cube) has 12 edges as well, and indeed an octahedron viewed from center has 6 independent directions (through each face center), but through each edge you could also define a reflection plane. More concretely, the symmetry group of a cube (or octahedron) is O_h of order 24 (excluding reflections, 24 proper rotations). Those include: - 90° rotations about axes through faces (three 4-fold axes), - 180° rotations about axes through edges (six 2-fold axes), - 120° rotations about axes through opposite vertices (four 3-fold axes).

The **12 edges** themselves correspond to 180° rotations (flip) about an axis through the centers of opposite edges – there are 6 such axes, each yields a π rotation symmetry (a half-turn). So one could identify 6 independent mirror transformations (180° flips) that send the cube to itself. If we include reflections (mirrors), each edge also lies in a plane of reflection symmetry (the plane that bisects that edge and the cube). There are indeed 6 mirror planes through opposite edges for a cube. So one can count either 6 or 12 depending on counting distinct transformations vs distinct oriented directions. It's a bit ambiguous but likely OIPK means 12 distinct *oriented directions* (e.g. 12 half-edges pointing out from the center). Each of those might define a direction along which the universe "mirrors" itself into a copy. This sounds like an *iterative symmetry*: you reflect the physical state across a plane or rotate it 180° around an axis, and perhaps that generates a copy of the structure. Doing it for all 12 edges' directions might generate multiple copies. If these copies then themselves undergo the process recursively, one gets a fractal of images.

This is reminiscent of a **kaleidoscope**. In a 3D kaleidoscope, if you had mirrors arranged in the shape of a rhombic dodecahedron (the polyhedron with 12 rhombic faces, which is actually the Voronoi cell of a cubic lattice), placing an object inside would produce multiple reflected images. A specific example: a "holoscope" built from 12 mirrors arranged as a rhombic dodecahedron produces an illusion of floating polyhedra [22]. Each mirror face yields reflections, and due to geometry, one can see recursively smaller images. This artistic device is essentially implementing the 12-plane reflection symmetry (each rhombic face corresponds to reflecting across that plane). For infinite reflections, if the angles between mirrors are rational fractions of π, one either gets a finite group or an infinite tiling. For the cube/octahedron, the angles lead to a *finite* group (24 rotations, and including mirrors 48 operations). So in a *purely symmetric ideal*, the reflections eventually produce repeats (after a finite number of operations, an image overlaps an existing one). That means it's not a true fractal but a set of at most 24 images. However, OIPK says "recursive mirroring" – implying self-similarity at smaller scales, which suggests something beyond the finite symmetry: likely a scaling factor is involved too (each mirror operation might also shrink the copy). This would turn the symmetry group into an *infinite self-similarity group*. A combination of reflections and uniform scaling by a factor (say $\lambda < 1$) yields an **iterated function system** (IFS) that can produce a fractal set. For example, consider 12 contractive maps M_i: each M_i could be "rotate space 180° about axis through edge i *and* scale by factor s towards the center". Applying all 12 to any shape and iterating yields a self-similar fractal (provided $12 s^3 < 1$ to ensure overall contraction in volume – here s is linear scale, so $12 s^D = 1$ with D dimension = 3 might define the fractal dimension as $D = \frac{\ln(1/12)}{\ln s}$). If one chooses s such that $12 s^3 = 1$, the fixed set would in fact be a space-filling solid (dimension 3) – trivial. For a fractal, s must be smaller. Suppose each iteration shrinks linear dimensions by $1/2$ (i.e. $s=0.5$). Then after one generation you have 12 half-size copies inside. They would occupy a fraction $12(1/2)^3 = 12/8 = 1.5$ *of the original volume – that actually exceeds 1, so that's not a contraction overall (would fill space redundantly). Try* $s = 1/3$: *then* $12(1/3)^3 = 12/27 \approx 0.444$, so it's a contraction mapping (volume reduces to 44%). Repeating infinitely yields a fractal of dimension D solving $12(1/3)^D = 1$ ⇒ $12\,3^{-D}=1$ ⇒ $3^{-D}=1/12$ ⇒ $-D \ln 3 = \ln(1/12)$ ⇒ $D = \frac{\ln(12)}{\ln(3)} \approx \frac{2.4849}{1.0986} \approx 2.262$. So dimension ~2.26 (just as an example). If one chooses $s=1/2$, $D = \ln(12)/\ln(2) = \frac{\ln 12}{\ln 2} = \log_2(12) \approx 3.585$. Many such choices possible. The fractal would visually look like a foam of smaller and smaller cubes or octahedra. This could be OIPK's spatial structure: space might not be smooth on all scales but have a nested self-similarity of structure at decreasing scales (perhaps related to a foam-like spacetime at Planck scale, reminiscent of quantum gravity notions). It's speculative but mathematically a well-defined possibility. Interestingly, **Penrose tilings** and quasicrystals often involve inflation/deflation symmetries (scale and rotate tiles). The 12-fold quasicrystal is one example (with scale factors related to $\sqrt{2}+1$ for dodecagonal patterns). OIPK's 12-direction fractal could be seen as a 3D quasicrystal if the scale factors are algebraic numbers – but we lack detail, so we keep it general.

1.5 Connection to Coxeter Groups: The reflections in a cube's symmetry are part of the **Coxeter group** $[4,3,2]$. If one were to allow infinite tilings, one might consider the group $[4,3,2]^+$ in hyperbolic space which leads to fractal limit sets (this ties into AdS tessellations etc., but that's deep group theory). In known cosmological chaotic models, like the Mixmaster (Bianchi IX universe), the evolution of the universe's anisotropy near the singularity can be described by a billiard on a triangular table in hyperbolic space, whose reflections correspond to the Weyl group of D_3 (which is isomorphic to the symmetry of an equilateral triangle). For Bianchi IX, this leads to a fractal behavior in time (the sequence of Kasner exponents has a fractal continued-fraction pattern) [75] [76]. In fact, **Mixmaster chaos** can be seen as the universe "mirroring" its contraction axis among the three spatial directions repeatedly. That yields an infinite sequence of Kasner epochs – a kind of self-similar (but chaotic) time fractal. OIPK's mirrors are in space rather than time, but it's intriguing that **billiard reflections** appear in both contexts. There is even an E8 related conjecture: the infinite reflections of Bianchi IX are associated with the E_{10} or E_8 hyperbolic Coxeter diagram [73] (though that's more speculative in literature). The mention of $E8$ in the directives might hint at this connection: E_8 plays a role in quasicrystal construction (the 8D E_8 lattice projects to 4D icosahedral structures) and in billiards of supergravity. While it's far-fetched to tie that directly to OIPK with current info, the presence of a *discrete symmetry with many mirrors* does resonate with Coxeter reflection theory. If OIPK spacetime has a discrete self-similarity, it might be described by a **Coxeter group with an affine extension** (to include scaling). For instance, the group of the cubic lattice extended with a scale could form an *affine Coxeter group* whose limit set is a fractal. Without digressing too much: mathematically, one could attempt to model the OIPK fractal with an affine Coxeter diagram for A_3 (cube), which likely yields a Sierpinski-like lattice structure. This is an avenue to explore if one were to rigorously define the fractal geometry.

1.6 Summary: In mathematical terms, validating OIPK involves showing a self-consistent solution to Einstein's field equations that encompasses: - **An anisotropic closed cosmology** with a bounce (expansion→contraction) – requiring either special matter (exotic pressure) or quantum gravity modifications to circumvent singularities. - **A discrete 12-fold symmetry** in the spatial sections (likely achieved if initial conditions imprint a cubic symmetry and matter respects it). - **Self-similarity** under that symmetry at smaller scales, which could be realized via an iterated function system or fractal distribution of matter/curvature. In practice, one might need scale-free (perhaps fractal) matter distributions – e.g., a fractal network of domain walls or strings that reproduce at smaller scales. This ventures beyond standard cosmology (which usually assumes a continuum fluid).

While constructing an exact OIPK solution is non-trivial, one can piece together plausibility from known components: **Kantowski–Sachs metrics** for the overall anisotropy, **mixmaster dynamics** for possible chaotic/fractal oscillations, and **holographic screens** for interpreting 2D slices as information carriers. The mathematical tools needed include solving ODEs for $a(\tau), b(\tau)$ and analyzing discrete symmetry transformations. Given the complexity, a full analytic solution likely isn't known – but a strategy could be: 1. Assume a form for matter (e.g., a scalar field with potential tuned to produce a turnaround at finite volume – similar to scalar-tensor bouncing cosmologies). 2. Impose octahedral symmetry in initial perturbations (so that as universe evolves, structures form aligned with those 12 directions). 3. Numerically integrate and see if fractal patterns emerge (e.g., does density clump in self-similar way along those axes?).

This would be a significant research project in itself, but from the foundations perspective, **no fundamental law of geometry prevents a cube-symmetric, anisotropic cosmology with a bounce.** It is a solution type allowed by Einstein's equations (with appropriate source). The addition of **consciousness** doesn't enter the math directly – except one could imagine that the "integration of slices" might impose a global constraint (like a consistency condition maybe analogous to a Hamiltonian constraint linking $\partial/\partial t$ and $\partial/\partial \tau$ evolutions, as earlier considered). If one were to incorporate that, one might introduce a potential function whose stationary condition yields a coupling of $\partial^2/\partial t \partial \tau$ terms. This is highly speculative since consciousness isn't in classical GR. One might instead think of it this way: OIPK's separation of dynamics means we could write the action as $I = I_{\text{vertical}}[\tau] + I_{\text{horizontal}}[t]$ to first approximation, and perhaps a coupling term $I_{\text{int}}[\tau,t]$. Variation w.rt fields could produce a mixed partial differential equation. For example, a **scalar field** $\Phi(x,y,z,t)$ might satisfy an equation $\partial^2 \Phi/\partial t^2 + f(\tau)\partial^2 \Phi/\partial \tau^2 = \nabla^2 \Phi$ if the background metric coupling yields such. In that fictitious equation, mixing of t and τ derivatives could occur for perturbations. This might be needed if one tries to quantize the system or account for feedback between fast (light) and slow (implosion) dynamics. Without a concrete physical reason (like two-time physics gauge symmetry as Bars uses [1]), we tread on thin ice. Given our scope, we acknowledge this as an open mathematical structure: **a spacetime with two time-like evolution parameters** can be made consistent only if constraints relate them (to avoid overcounting degrees of freedom and ensure determinism [3]). In OIPK, presumably one parameter (t) is an actual physical time, and the other (τ) is like a label for the background solution – so it might be better thought of as a *background parameter* rather than a second physical time. That interpretation skirts the issues Tegmark raised about predictability with two times [3] , by effectively not letting τ vary freely as an independent axis – instead, the universe has a single timeline but is structured vertically in a special way.

In conclusion of the mathematical foundation: we have formulated OIPK's spacetime metric, identified conditions for its evolution, and outlined how its unusual symmetries might manifest. The framework is **geometrically rich** – touching on anisotropic cosmologies, discrete symmetry groups, and fractal geometry. Each of these is individually sound in mathematical physics; the challenge is uniting them in one consistent solution. The viability of OIPK rests on finding an appropriate stress-energy to drive the desired dynamics (likely requiring new physics, as classical dust or radiation won't bounce) and ensuring that the large discrete symmetry doesn't conflict with observed isotropy (implying either a special epoch or subtle implementation like domain averaging). These are hurdles for Sections 5 and 10 (Challenges and Experiments) to examine. But at least at the level of *in principle*, **general relativity is flexible enough to accommodate a diamond-shaped, multiply symmetric spacetime** – it's a matter of finding the right content and initial conditions, and possibly invoking quantum gravity at the crucial moments (the vertices).

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